

3D directional tuning in the orofacial sensorimotor cortex during natural feeding and drinking

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This **useful** study characterises motor and somatosensory cortex neural activity during naturalistic eating and drinking tongue movement in nonhuman primates. The data, which include both electrophysiology and nerve block manipulations, will be of value to neuroscientists and neural engineers interested in tongue use. Although the current analyses provide a **solid** description of single neuron activity in these areas, both the population level analyses and the characterisation of activity changes following nerve block could be improved.

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Abstract

Directional tongue movements are crucial for feeding and speech, ensuring proper food positioning for chewing and swallowing, as well as accurate sound production. While directional tuning in the arm region of the sensorimotor cortex during reaching tasks is well-studied, little is known about how 3D tongue direction is encoded in the orofacial sensorimotor cortex (OSMCx) during natural behaviors. Understanding this neural representation has important implications for rehabilitating individuals with orolingual dysfunctions. This study examines the directional tuning and population dynamics in OSMCx during naturalistic feeding and drinking, and how these are affected by sensory loss. Using biplanar videoradiography, we tracked implanted tongue markers in behaving rhesus macaques (*Macaca mulatta*) and simultaneously recorded 3D positional data with spiking activity from chronically implanted microelectrode arrays in primary motor (M1o) and somatosensory (S1o) areas of the orofacial cortex. In some sessions, tasks were preceded by bilateral nerve block injections to the sensory branches of the trigeminal nerve. Modulation to 3D tongue direction during feeding and drinking was found in most M1o and S1o neurons. Directional information in both individual- and population-level was higher in feeding and was more robust in M1o. Following sensory loss, alterations in tongue kinematics were accompanied by changes in directional information in M1o and S1o, manifesting as modifications in both individual neuron tuning characteristics and the broader dynamics of population-level neural activity. Overall, this study advances our understanding of how

OSMCx contributes to complex, coordinated control of naturalistic tongue movements. It expands our current knowledge of orofacial control to three dimensions and demonstrates the specificity and adaptability of population activity in MIO and SIO in response to different behavioral contexts.

Introduction

Motor and somatosensory cortical neurons modulate their spiking activity based on movement direction as seen in arm reaching tasks (Georgopoulos et al., 1988 [↗](#); Schwartz et al., 1988 [↗](#); Prud'homme and Kalaska, 1994 [↗](#)) and orofacial behaviors. In the primary motor (MIO) and primary somatosensory (SIO) areas of the orofacial sensorimotor cortex (OSMCx), neurons encode the direction of voluntary tongue protrusion (Murray and Sessle, 1992 [↗](#); Lin et al., 1994 [↗](#)) and semi-automatic tongue movements in chewing and swallowing (Sessle et al., 2005b [↗](#)). Extensive research has explored how the arm region of the sensorimotor cortex encodes movement direction (Ajemian et al., 2000 [↗](#); Georgopoulos et al., 2007 [↗](#); Churchland et al., 2012 [↗](#); Lillicrap and Scott, 2013 [↗](#)). Since the tongue is enclosed within the oral cavity and thus hidden from view, it has proved difficult to study the neuromechanical processes underlying directional tongue movements that are essential for these behaviors (Hiimae and Palmer, 2003 [↗](#)). Thus, considerably less is known about how 3D tongue direction is encoded in the OSMCx and the role of tactile sensation (Bach-y-Rita et al., 1998 [↗](#); Lamm et al., 2005 [↗](#); Lozano et al., 2009 [↗](#)) during natural feeding and drinking. This knowledge has important implications for improving evaluation and treatment strategies for individuals with sensorimotor dysfunctions (Takizawa et al., 2016 [↗](#); Avivi-Arber and Sessle, 2017 [↗](#)).

The OSMCx plays an important role in the coordination of complex tongue movements. Seminal studies on the directional tuning properties of OSMCx neurons by Sessle and colleagues employed varying locations of spouts that delivered a juice reward to elicit directional tongue protrusions without tracking tongue movements. A later study incorporated tracking of 2D tongue movements using videofluoroscopy during voluntary directional protrusions (Arce et al., 2013 [↗](#)), but the tongue trajectories were not used to study directional tuning. In all these prior studies, primates have been trained to interact with a computer display to elicit a tongue protrusion to a specific direction on cue. There is a knowledge gap on how spiking activity in the OSMCx relates to tongue movements during natural behaviors. With the development of biplanar video-radiography (Brainerd et al., 2010 [↗](#)), it is now possible to track these 3D tongue movements within the oral cavity at a high temporal and spatial resolution (Montuelle et al., 2020 [↗](#); Feilich et al., 2021 [↗](#)). By simultaneous recording of precise tongue movements and spiking activity, we have shown recently that tongue position and shape can be accurately decoded from OSMCx during feeding (Laurence-Chasen et al., 2023 [↗](#)).

Our study investigated how OSMCx encodes and decodes tongue direction during untrained feeding and drinking, comparing activity across multiple cortical regions. Given the importance of oral somatosensation in tongue positioning and bolus control during chewing and swallowing (Smith and Cutrer, 2011 [↗](#)), we also examined its role in sensorimotor control by selectively blocking oral tactile sensation.

Methods

Subjects

Experiments were performed on two adult male rhesus macaques (*Macaca mulatta*, 9–10 kg, ages 8 and 9 years) in the University of Chicago XROMM Facility. This sample size was chosen based on precedent in the field of non-human primate motor neuroscience. All protocols were approved by the University of Chicago Animal Care and Use Committee and complied with the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

The subjects were seated in a standard primate chair and head-fixed to keep their head position constant during feeding and drinking trials. Each trial lasted 10 seconds. In a feeding trial, a piece of food (grape, gummy bear, pasta) of roughly the same size was presented directly to the animals' mouth using a stylus. In a drinking trial, juice was delivered through one of three spouts positioned in front of the subject (**Fig. 1A** [↗](#)).

Cranial Nerve V anesthesia

For some sessions, these behavioral tasks were preceded by nerve block injections (0.25% Bupivacaine HCL and Epinephrine 1:200,000, 0.25 mL/injection site) to the sensory branches of bilateral trigeminal nerves (lingual, inferior alveolar, buccal, palatine) to eliminate oral tactile sensation locally and temporarily. The nerve block was administered while the subjects were under sedation, and all data were collected within 90 minutes of the nerve block. Each monkey served as its own control, with nerve block feeding data collection sessions taking place either a day before or a day after the associated control session. Nerve block drinking data collection was performed immediately following the control drinking session. Multiple datasets (40–60 trials) were collected for both subjects across multiple days. However, due to the complex and time-consuming nature of processing integrated XROMM and neural data, one session per subject, behavior, and condition was used for this study. Thus, we analyzed a total of 8 datasets.

Video-radiography

Prior to data collection, the animals were implanted with spherical tantalum beads (1-mm diameter) in the cranium, mandible, and the tongue, from the tip to the region of the circumvallate papillae. During feeding or drinking, the movement of these markers was recorded using high-resolution (200 Hz, <0.1 mm) biplanar video-radiography collected with Xcitex ProCapture version 1.0.3.6. The 3D positional data was obtained following the previously described X-ray Reconstruction of Moving Morphology (XROMM) workflow ([Laurence-Chasen et al., 2020](#) [↗](#)) incorporating the use of XMALab ([Knörlein et al., 2016](#) [↗](#)) and machine learning using DeepLabCut ([Mathis et al., 2018](#) [↗](#)) to reconstruct the kinematic data. The x , y , z values of the markers were then smoothed with a 30 Hz low-pass Butterworth filter and transformed into a cranial coordinate space with the origin fixed at the posterior nasal spine. Gape cycles within each feeding sequence were manually identified and categorized by cycle type (manipulation, stage 1 transport, chew, stage 2 transport, or swallow).

Electrophysiology

Under general anesthesia, a microelectrode array was chronically implanted in four areas of the left hemisphere (Supplementary file 1, **Fig. 1** [↗](#)): rostral MIO (96-electrode Utah array; electrode length: 1.5 mm), caudal MIO (32-electrode Floating microelectrode array (FMA), electrode length: 3.0–4.5 mm), area 1/2 (96-electrode Utah array, electrode length: 1.0 mm), and area 3a/3b (32-electrode FMA, electrode length: 4.0–8.7 mm). The neural data was recorded using Grapevine Neural Interface Processor (Ripple Neuro, Salt Lake City, UT). Signals were amplified and bandpass filtered between 0.1 Hz and 7.5 kHz and recorded digitally (16-bit) at 30 kHz per channel. Only waveforms (1.7 ms in duration; 48 sample time points per waveform) that crossed a threshold were stored and offline spike sorted (Offline Sorter, Plexon, Dallas, TX) to remove noise and to

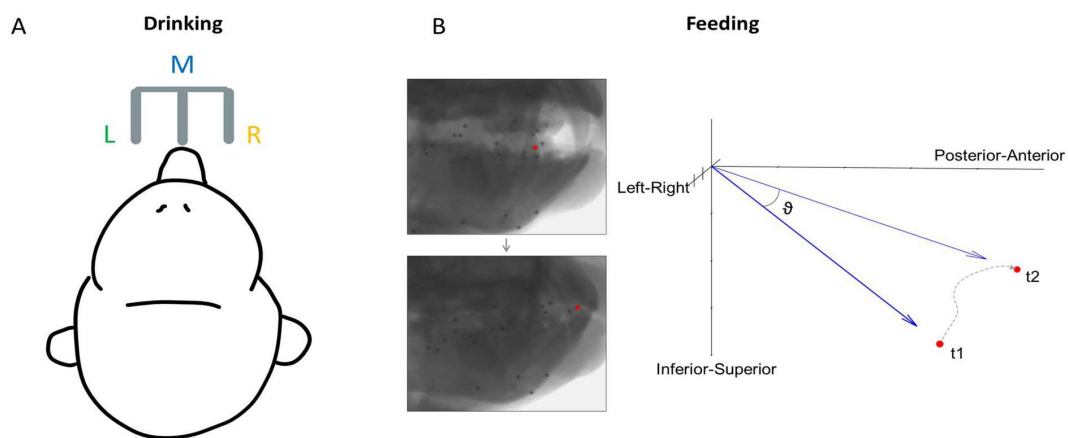


Figure 1.

Direction of tongue motion in each behavioral task.

(A) Schematic of the location of three spouts, left (L), middle (M), and right (R), for the drinking task. Tongue direction was categorized based on spout location. **(B)** Calculation of 3D tongue direction during feeding. θ is the instantaneous 3D direction of the tongue tip over a 100 ms interval between its positions at t_1 and t_2 , where $t_1 = 0$ and $t_2 = t_1 + 100$. The dotted line shows the actual trajectory during this interval.

isolate individual neurons. Neurons recorded during control feeding sessions are the same as those previously reported on in [Laurence-Chasen et al., 2023](#). The channel name assigned to each recorded neuron was kept consistent between control and nerve block data for comparison.

Data analysis

3D kinematics

3D tongue kinematics were recorded simultaneously with the neural data in all behavioral sessions. All data analyses were performed in MATLAB 2022b (MathWorks, Natick, MA). For feeding, the instantaneous 3D direction of the tongue tip marker for every 100 ms throughout each gape cycle was calculated as:

$$3D\ angle, \theta = \tan^{-1}(\|v_1 \times v_2\| / v_1 \cdot v_2) \quad (1)$$

Where v_1 is the x, y, z position at the start of each 100-ms interval and v_2 is the position at the end (**Fig. 1B**). These directions were then categorized based on whether the movement was negative or positive relative to the horizontal plane (Left/Right), the sagittal plane (Inferior/Superior), and the x axis (Posterior/Anterior). This resulted in eight directions: AntSupL, AntSupR, AntInfL, AntInfR, PostSupL, PostSupR, PostInfL, and PostInfR. An equal number of 100-ms intervals from each of these directions was sampled to eliminate the possible effect of different distributions of kinematics across datasets, and spike data during each interval was used for neural analysis. For comparison with the drinking task, the sign was determined relative to the horizontal plane, with rightward tongue movement being positive. This is also the plane of motion which has been the least studied. These left-right directions were categorized into six 10°-bins with a total range of -30° to 30°, which encapsulated most of the observed distribution of directions in each subject. Lingual yaw (transverse rotation) and pitch (elevation/depression) were also calculated to compare tuning across the lateral and vertical components of tongue direction (Supplementary file 2, **Equation 2**). For drinking, the direction was determined by which of the three spouts juice was dispensed from during each lick. The spiking activity used for neural analysis of the drinking task was from intervals of ± 250 ms around each minimum protrusion of the tongue. As 100 ms was not sufficient to capture the full range of tongue motion during each drinking cycle, the length of time used was increased to allow a clear distinction between the three directions. The period of 250 ms spans about 75% of the average lick length from minimum to maximum protrusion of the tongue.

Kinematic performance for feeding was determined by the spread of tongue directions observed across trials. For drinking trials, performance was determined by the variance of endpoint positions as well as by the proportion of “failed” cycles, where the monkey missed the correct spout location with their tongue tip. The difference between control and nerve block performance was evaluated using a two-tailed t-test and f-test.

Directional tuning of single neurons

Tongue directions were subsequently compared with the firing rates of individual neurons across cortical areas. To determine if neurons were directionally modulated, we used a bootstrap procedure ([Arce et al., 2010](#)): we resampled the firing rates from an equal number of trials in each direction with replacement 1000 times and computed 95% confidence intervals from the resulting distribution to test whether the mean ranks are the same across directions. The proportions of neurons that were found to be directionally tuned were compared across groups using a chi-square test. Due to limited neuron counts in some cortical regions, we combined rM1 and cM1 recordings as M1o, and areas 1/2 and 3a/3b as S1o for subsequent analyses (Supplementary file 1, Table 1). Then, multiple linear regression was used to determine if the firing of each neuron fit the cosine tuning function that has been previously described for the arm

area of the motor cortex (Schwartz et al., 1988 [↗](#)). To accomplish this, the directional components of a unit vector representing each group of directions were calculated. For neurons that fit the tuning function, a preferred direction (PD) in 3D space was estimated. These PDs are distributed around a unit sphere, with the origin representing the start of the movement. The directional index was calculated as a measure of the depth of directional tuning. To determine PDs for the drinking task, we resampled the original distribution of firing rates with replacement for each direction and calculated the direction for which a neuron exhibited its maximal firing rate over 1000 bootstrap samples. Similarly, a PD across the left-right feeding directions was determined for comparison. Circular concentration (k-test) to compare distributions of PDs during feeding and polar plot generation were performed using the CircStat MATLAB toolbox (Berens, 2009 [↗](#)). For drinking, distributions of PDs were compared using a chi-square test. We analyzed the trial-by-trial variability of neuronal activity using the Fano factor, which was computed as the spike-count variance divided by spike-count mean within each session. The Fano Factor was calculated separately for each subject, task, and cortical region. For analysis across sessions, we used the mean-matched Fano factor (Churchland et al., 2010 [↗](#)).

Factor Analysis of population activity

We used Factor Analysis (FA), a linear dimensionality reduction method, to obtain latent trajectories of spiking activity and compare population responses to different directions across trials. FA defined as:

$$y \sim N(\mu, CC' + R) \quad (2)$$

where y is the spike counts from n neurons, μ is the mean spike counts from n neurons, C ($m \times n$) is the loading matrix mapping m latent factors to the spike counts of n neurons, and R ($n \times n$) represents the unexplained variance of independent neurons (Santhanam et al., 2009 [↗](#); Horrocks et al., 2024 [↗](#)). The parameters μ , C and R were estimated with expectation-maximization using DataHigh Matlab toolbox (Cowley et al., 2013 [↗](#)). The latent factor space obtained using FA represents the shared population variance of neurons. To obtain latent factors of shared population activity, we binned the spike times of each neuron into 10 ms bins (10 bins for feeding; 50 bins for drinking). Neurons with a mean firing rate < 1.0 spike/s were excluded and the resultant vectors were smoothed using a Gaussian kernel with a 10 ms standard deviation (SD). To determine the dimensionality of the latent variable, we used 3-fold cross-validation to find the value of m which maximized the likelihood of the data. We then obtained an FA model by fitting an m -dimensional latent factor model. FA was performed using trial-averaged data to examine the direction-relevant latent factor responses (i.e., neural population trajectories).

We quantified directional differences in population activity by calculating the Euclidean distance over m latent factors) between trial-averaged neural population trajectories for each unique direction pair (drinking = 3 pairs; feeding = 28 pairs). This analysis was performed for every 10 ms bin throughout each trial. To assess differences between experimental conditions (control vs. nerve block) or cortical region (Mlo vs. SIo), we applied two-sample t-test to the mean inter-trajectory distances across all direction pairs. We further characterized the neural space spanned by population activity by measuring the cumulative Euclidean distance travelled by trajectories from start to end of a trial.

To control for potential sampling biases, we implemented two critical validation procedures. First, we addressed the varying trial counts across directions in the feeding task by performing Factor Analysis with standardized samples ($N = 80$ trials per direction) through random subsampling repeated 10 times. We then compared the cumulative explained variance between the full and subsampled datasets. Second, we controlled for population size differences by subsampling Mlo and SIo neurons to equivalent counts, enabling unbiased comparison of Factor Analysis results between cortical regions.

To compare behaviors, we performed FA on only caudal MIO neurons that remained stable across both feeding and drinking recording sessions ($N = 20$). These neurons were determined through a stability test (Dickey et al., 2009), which compared the average waveform and interspike interval for both datasets. We analyzed two subsets of kinematic data: data collected within 200 ms surrounding minimum gape, and data from trials where the 3D angle measured 100 ms after minimum tongue protrusion was between -5 and $+5$ degrees.

Decoding tongue direction

The ability to predict tongue direction from spiking activity of MIO and SIO neurons was evaluated using a K-nearest neighbor (KNN) classifier. The Euclidean distance was used to identify nearest neighbors, and the number of nearest neighbors used was $K = 7$. This K value was determined after testing different Ks which yielded comparable results. The feature was the firing rate of each neuron over each trial: every 100 ms throughout feeding sequence, or 100 ms centered at minimum tongue protrusion during drinking. As a more direct comparison to the drinking, feeding directions were split into three groups representing left, middle, and right movement directions. The decoder was trained on 80% of trials and tested on the remaining 20%, then decoder performance was determined by the percentage of test trials where the direction of movement was correctly decoded from the neural data. We ran 100 iterations of the classifier using a different set of randomly selected training and test trials then calculated the average performance. The same sets of training and test trials were used for decoding from simultaneously recorded MIO and SIO data. However, our recorded populations were of variable sizes, and decoding performance was found to be related to the number of neurons in the ensemble. Because the smallest population of neurons we recorded was 28, we selected 28 random neurons from the larger populations for each iteration. Based on the positive relationship between population size and decoding accuracy, we expect that performance would increase with more neurons. These results will show whether tongue direction can be decoded from a very small number of neurons. We fit a linear regression model with interactions to compare decoding performance across the other variables in the experiment. To determine if a mixed population of MIO and SIO neurons performs better than the pure populations, we started with the full MIO population and systematically replaced 25 MIO neurons with an equal number of SIO neurons. We repeated this replacement over 100 iterations, each with a different random selection of neurons, and decoded tongue direction using the KNN classifier. We compared the average decoding performance of these slightly different mixed populations to the baseline of running the decoder with the full MIO population.

We also decoded tongue direction using a long short-term memory (LSTM) network (Hochreiter and Schmidhuber, 1997; Glaser et al., 2020; Laurence-Chasen et al., 2023; Hahn and Arce-McShane, 2024) implemented in MATLAB's Deep Learning Toolbox. For the feeding task, we analyzed 3D tongue movement direction (relative to the sagittal plane) at 100 ms intervals. For the drinking task, we categorized tongue direction every 500 ms as either left (-45°), middle (0°), or right (45°). For each neural population, we created a 2D array containing spike counts for these time intervals (neurons $\times$ intervals). We randomly selected five groups of 28 neurons with replacement from each population. Using 5-fold cross-validation, we trained an LSTM network (400 hidden units, 50 training epochs) on 85% of the intervals. The network was then tested on the remaining 15% of neural data in a stepwise manner, producing a sequence of predicted tongue directions. We used mean R^2 to measure prediction accuracy.

Results

Previous research has investigated directional tuning of OSMCx neurons during trained tongueprotrusion tasks. Our study extends this work by investigating two natural, untrained behaviors: feeding and licking (“drinking”) from a spout. These behaviors provide distinct contexts

for studying tongue coordination - feeding allows unrestricted tongue movement, while drinking constrains movement direction based on spout location. This comparison enables us to examine how OSMCx coordinates tongue movements under different behavioral contexts.

Neuronal modulation patterns differ between MIO and SIO

Many neurons exhibited significant modulation of spiking activity to tongue direction (bootstrap, $p < 0.05$), though there were diverse patterns. **Figure 2** shows peri-event time histograms (PETHs) of two example neurons related to tongue movements during feeding and drinking. In feeding, both neurons showed complex oscillatory firing, with notable peaks between -0.05 s to 0.1 s for upward movements to the right (MIO, **Fig. 2A-left**) and posterior movements to the left (SIO, **Fig. 2A-right**). For drinking there was clear separation between different directions in the MIO neuron, with the left (green) exhibiting the highest activity (**Fig. 2B-left**), while SIO showed oscillatory patterns with less distinct separation between spout locations, but higher overall activity (**Fig. 2B-right**). Like the arm region, the tuning curves of directionally modulated MIO and SIO neurons fit the cosine tuning function (F -test, $p < 0.05$, feeding: MIO = 86%, SIO = 75%). **Figure 3A** maps a neuron's firing rate for tongue movements in left-right, inferior-superior, and posterior-anterior axes. Here, a MIO neuron is strongly tuned to posterior-anterior and inferior-superior directions, while remaining unresponsive to left-right movements during natural feeding. Many of the recorded neurons in each population behaved in a similar fashion, with peaks most frequently observed toward the anterior and superior directions. This observation was consistent with the tongue movements being most frequent in the Anterior Superior directions, followed by the Posterior Inferior (Figure 3 – figure supplement 1. The varying neuronal responses to tongue direction could not be attributed to variability in their firing, as the distribution of the Fano factor was similar across directions (Kruskal-Wallis, $p > 0.1$ for all except SIO control drinking $p = 0.06$).

The proportion of directionally tuned neurons was higher in the feeding vs. drinking task (Chi-square, $p < 0.05$, feeding: 72%, drinking: 66%) and differed significantly between MIO and SIO during the feeding task in both subjects (Chi-square, $p < 0.001$). In rostral and caudal MIO, 80% of neurons were modulated to 3D direction (bootstrap, $p < 0.05$, **Fig. 3B**, left), compared to 52% in areas 1/2 and 3a/3b. Notably, fewer MIO neurons showed directional tuning during swallows compared to chewing, while SIO neurons maintained consistent proportions (Figure 3 – figure supplement 2; Chi-square, MIO: $p < 0.05$, SIO: $p > 0.1$). During drinking, the proportion of directionally modulated neurons was more similar between regions (69% in MIO vs. 60% in SIO: Chi-square, $p > 0.05$, **Fig. 3B** right). Mean-matched Fano factor was significantly lower in MIO than SIO in both tasks (Wilcoxon rank sum test, $p < 0.001$). We considered that the difference in the directional tuning between the two behaviors could be due to the different time intervals used for each task since the period around minimum tongue protrusion in the drinking may contain more of the sensory inputs from the previous lick. However, when sampling spiking activity from an earlier period in feeding, the percentage of directionally tuned SIO neurons was still significantly lower than MIO (Chi-square, $p < 0.001$, data not shown). Shifting the period of the drinking trials to be after minimum protrusion also did not lead to a significant difference between MIO and SIO ($p > 0.1$).

Further analysis of the tongue's lateral (yaw) and vertical (pitch) components during feeding revealed additional insights. **Figure 4A** shows peak activity of a neuron in MIO and in SIO at varying degrees of pitch and yaw. Overall, more neurons responded to pitch than yaw (**Fig. 4B**), with MIO showing a higher proportion of neurons tuned to both components compared to SIO (Chi-square, yaw: $p < 0.08$, pitch: $p < 0.001$), consistent with our 3D direction results. MIO neurons exhibited sharper and narrower tuning curves than the broader tuning curves observed in SIO (**Fig. 4A**, see Supplementary file 2).

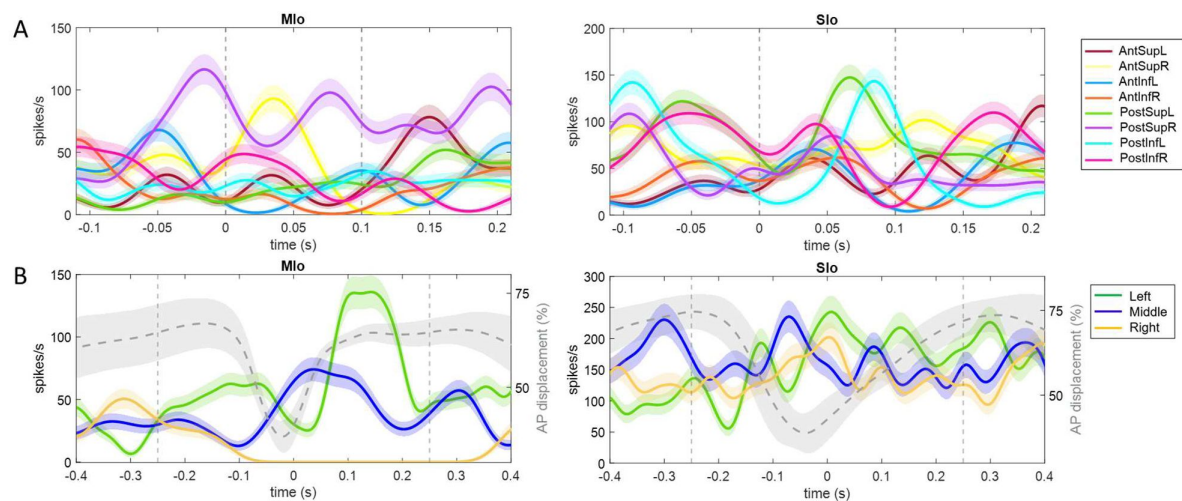


Figure 2.

Examples of single neuron activity in relation to tongue direction.

(A) Each peri-event time histogram (PETH and ± 1 SE, smoothed by a 25-ms Gaussian kernel) corresponds to spiking activity for a specific range of tongue direction for feeding trials. Dashed lines indicate 100-ms interval used for calculating the tongue direction. **(B)** PETHs for drinking trials with the same spout, centered at the point of minimum protrusion of the tongue (0-s). Percent tongue displacement along the anterior-posterior axis is shown in grey, with shaded area representing ± 1 SD. Vertical lines indicate 500-ms interval used for tuning analysis.

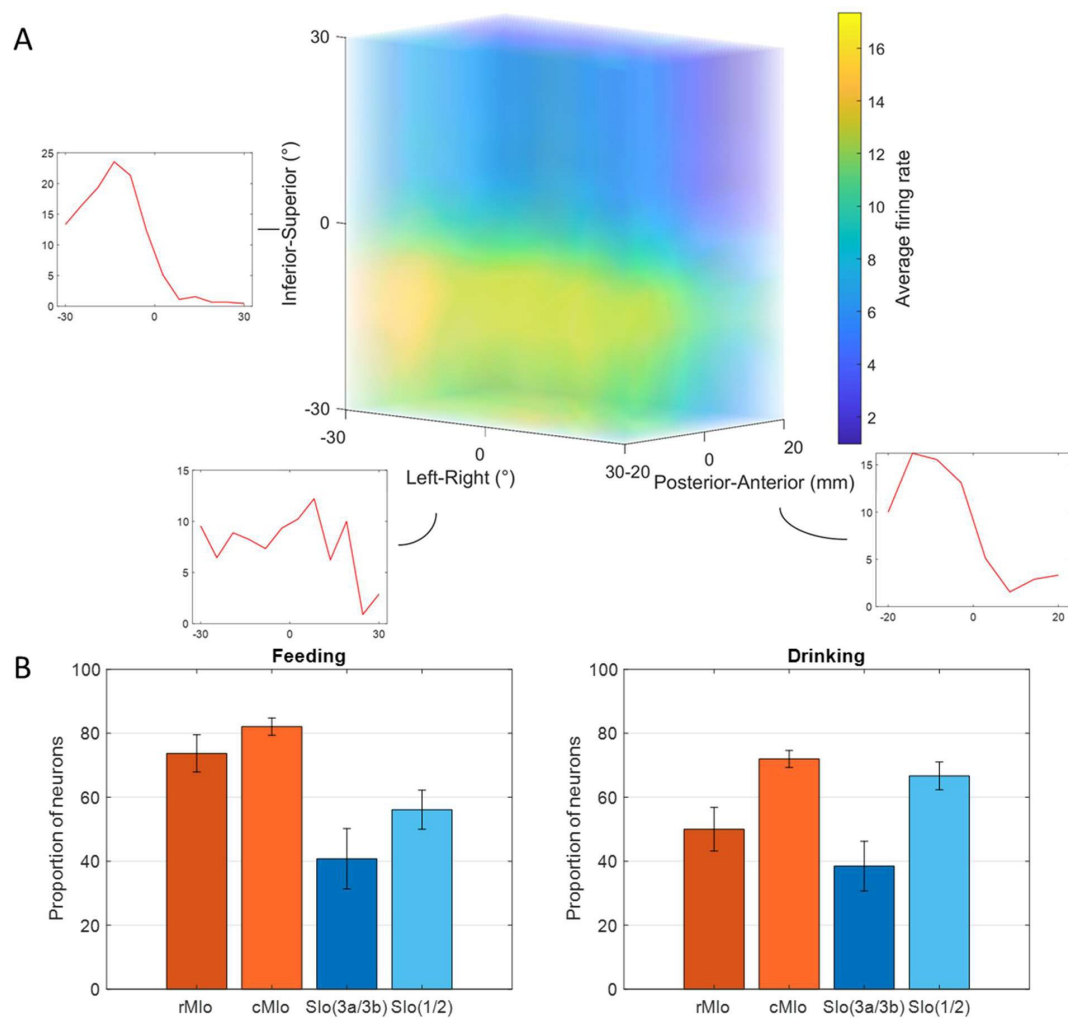


Figure 3.

Directional tuning of neurons during control tasks.

(A) 3D firing rate map of a neuron in MIO during feeding. Smaller inset plots are 1D tuning curves across each axis. **(B)** Percentage of neurons tuned to direction, combined for both subjects. Recordings were taken from four areas of the OSMCx: rMlo - rostral M1, cMlo - caudal M1, Slo(3a/3b) - area 3a/3b, and Slo(1/2) - area 1/2. Error bars represent ± 1 SE.

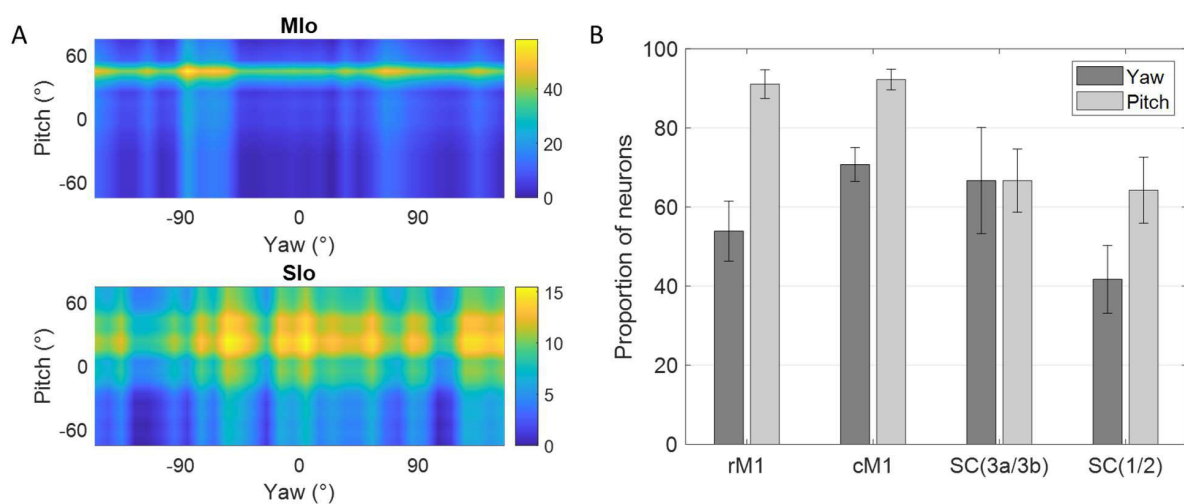


Figure 4.

Directional tuning to yaw and pitch during feeding.

(A) Firing rate maps of a neuron in Mlo and in Sio across yaw and pitch angles. Firing rates were averaged across all 100 ms feeding intervals within a 10° range. **(B)** Proportion of neurons tuned to yaw and pitch, combined for both subjects. Recordings were taken from four areas of the OSMCx: rM1 - rostral M1, cM1 - caudal M1, SC(3a/3b) - area 3a/3b, and SC(1/2) - area 1/2. Error bars represent ± 1 SE.

The peak of the PD distribution coincides with leftward tongue movements

The distribution of preferred directions provides insight into how tongue muscles are coordinated during movement. Intrinsic and extrinsic tongue muscles are involved in shaping the tongue (e.g., elongation, broadening) and positioning the tongue (e.g., protrusion/retraction, elevation/depression), respectively. These muscles receive bilateral motor innervation except for genioglossus. Straight tongue protrusion requires the balanced action of the right and left genioglossi while the lateral protrusion involves primarily the contralateral genioglossus. Given this unilateral innervation pattern, we hypothesized that left MIO/SIO neurons would preferentially respond to leftward tongue movements, corresponding to right genioglossus activation.

During feeding, MIO and SIO showed non-uniform distribution of preferred directions across a unit sphere (**Fig. 5**; Rayleigh test, $p < 0.001$) and similar directional indices (t-test, $p > 0.1$, mean $\pm$ 1 SD: MIO: 0.533 ± 0.3 , SIO: 0.604 ± 0.5). As hypothesized, most neuronal populations showed peaks in PD distributions toward leftward tongue movements, except in Monkey R's SIO (**Fig. 6A**). Similar results were found with the distributions of preferred yaw during feeding (Supplementary file 2, **Fig. 4**). While feeding showed comparable PD distributions between MIO and SIO in both subjects (circular k-test, $p > 0.1$), drinking revealed significant differences between regions in Monkey R (Chi-square, $p < 0.001$) but not Monkey Y ($p > 0.09$). Monkey Y maintained predominantly left-directed PDs across both tasks, while Monkey R showed more balanced left-right PDs during drinking, suggesting potential involvement of additional muscles beyond the right genioglossus.

Neural population trajectories differed based on task and cortical regions

We analyzed directional tuning at the population level using Factor Analysis (FA) on simultaneously recorded neurons to extract neural trajectories and identify patterns in their shared activity. In both tasks, neural trajectories of population activity in both MIO and SIO exhibited robust directional information; inter-trajectory distances of all unique direction pairs were significantly higher than zero (**Fig. 7**, t-test, $p < 0.001$, for all comparisons in both subjects and region). Notably in the feeding task, MIO and SIO showed smaller inter-trajectory distances between Anterior-Posterior paired trajectories during feeding (e.g., AntSupL-brown and PostSupL-green) compared to a greater separation between other directional pairs (**Fig. 7-top**, t-test, $p < 0.05$ for both monkeys). Paired Superior-Inferior directions, e.g., PostSupL-green and PostInfL-cyan, showed the largest separation, though this difference was not significant ($p > 0.1$). Cumulative explained variance for the first three factors was higher in feeding (MIO: 82%, SIO: 81%) than in drinking (MIO: 74%, SIO: 63%) when all neurons were used for the factor analysis (**Fig. 7**). Similar results were obtained when equal number of neurons were used (**Fig. 7-figure supplement 1**). To control for factors such as different neurons and kinematics that might influence the results, we performed factor analysis on stable neurons across both tasks using all trials (**Fig. 7-figure supplement 2A**) and using trials with similar kinematics (**Fig. 7-figure supplement 2B**). While the general shape of the population trajectory was preserved across tasks, the inter-trajectory distance between them was significant (t-test, $p < 0.001$, mean $\pm$ 1 SD: 0.6786 ± 0.0424). Moreover, the MIO trajectory length was longer in feeding (feeding: 0.8954, drinking: 0.6038) despite similar firing rates across tasks (t-test, $p > 0.05$) and longer tongue displacement in drinking ($p < 0.001$, mean $\pm$ 1 SD: feeding: 27.5 ± 9.8 , drinking: 47.6 ± 11). When trials were limited to those with similar directional angles (± 5 degrees), the difference in trajectory length was no longer observed but the inter-trajectory distance between tasks remained significantly different (t-test, $p < 0.001$, mean $\pm$ 1 SD: 0.6967 ± 0.0793).

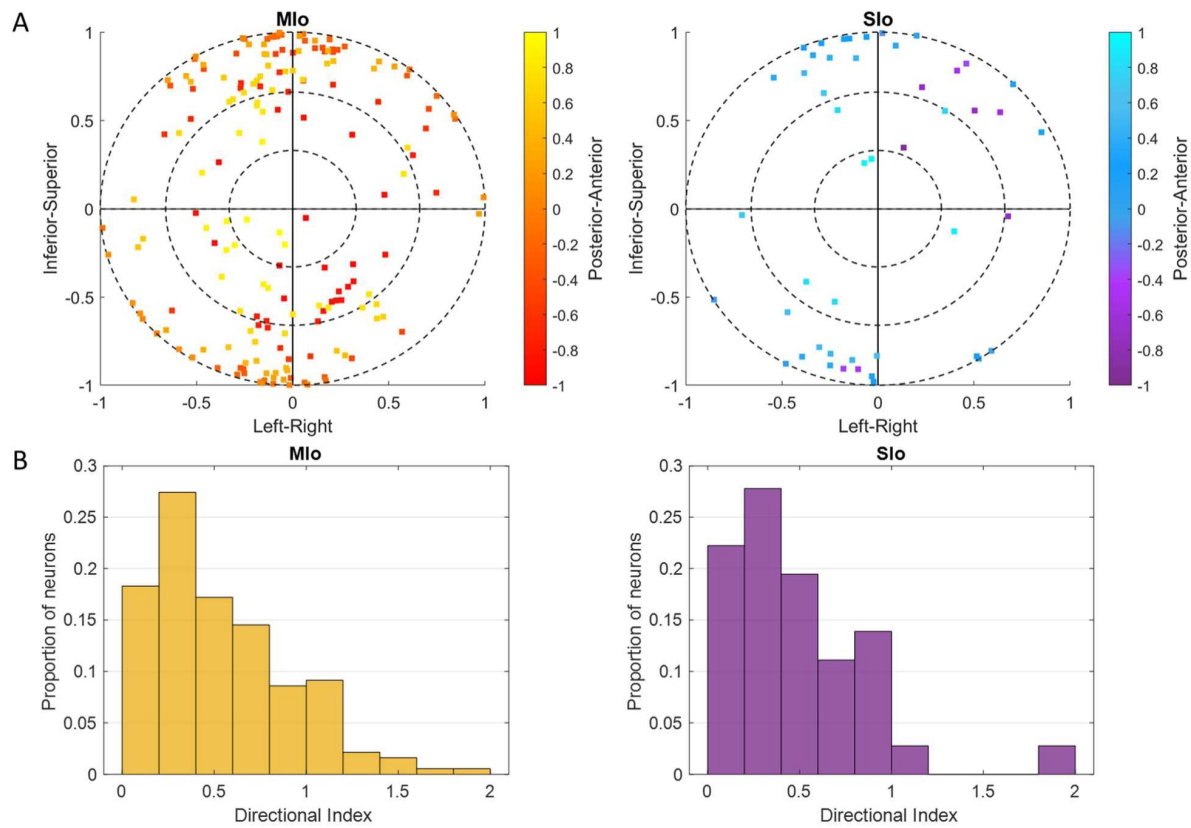


Figure 5.

Cosine tuning of MIO and SIO neurons.

(A) Distribution of 3D preferred directions in unit sphere for neurons that fit the tuning function during feeding, combined for both subjects. The origin represents the start of a movement. Color bar represents posterior-anterior axis. **(B)** Distribution of the index for the depth of directional tuning, combined for both subjects.

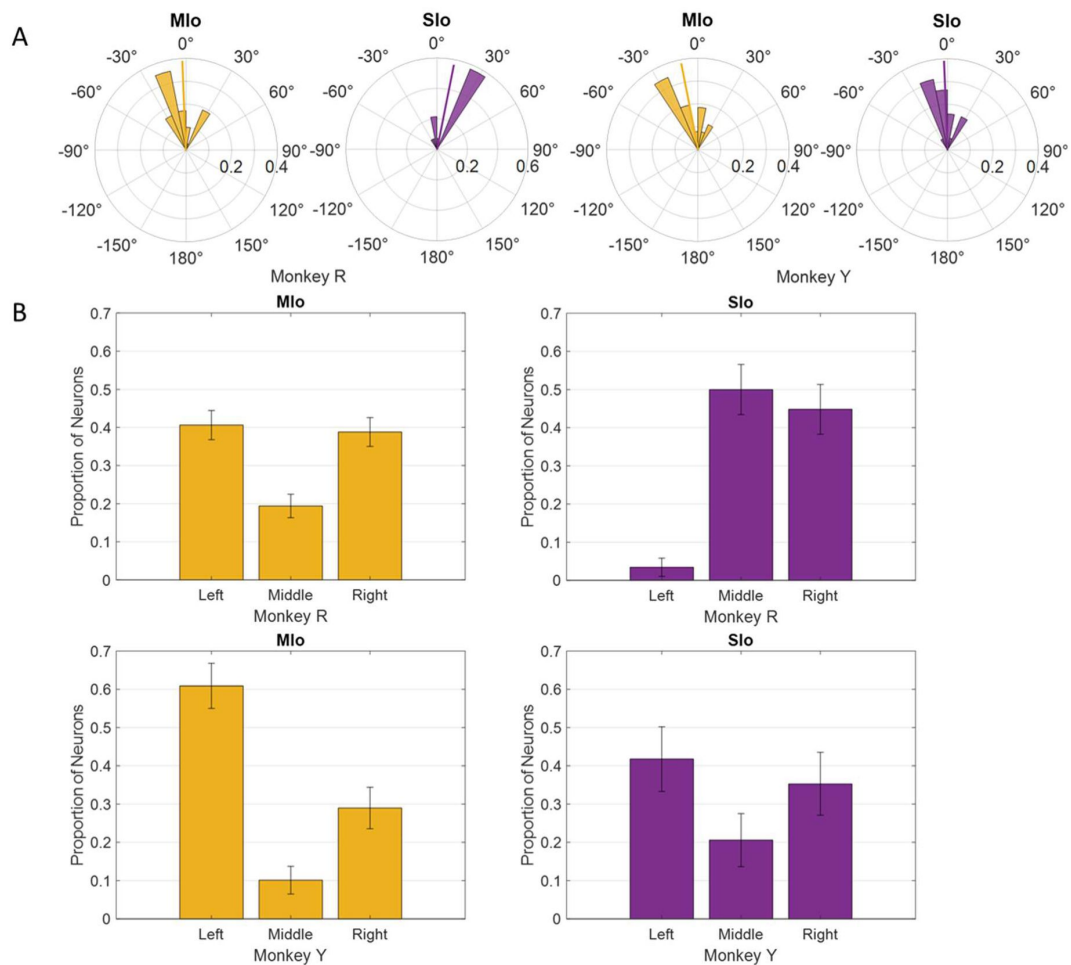


Figure 6.

Distribution of PDs in MIO (yellow) and SIO (purple) neurons during control feeding (A) and drinking (B).

For the feeding task, polar plots are split into 10° bins with thick colored lines representing the mean PD. For the drinking task, error bars represent ± 1 SE.

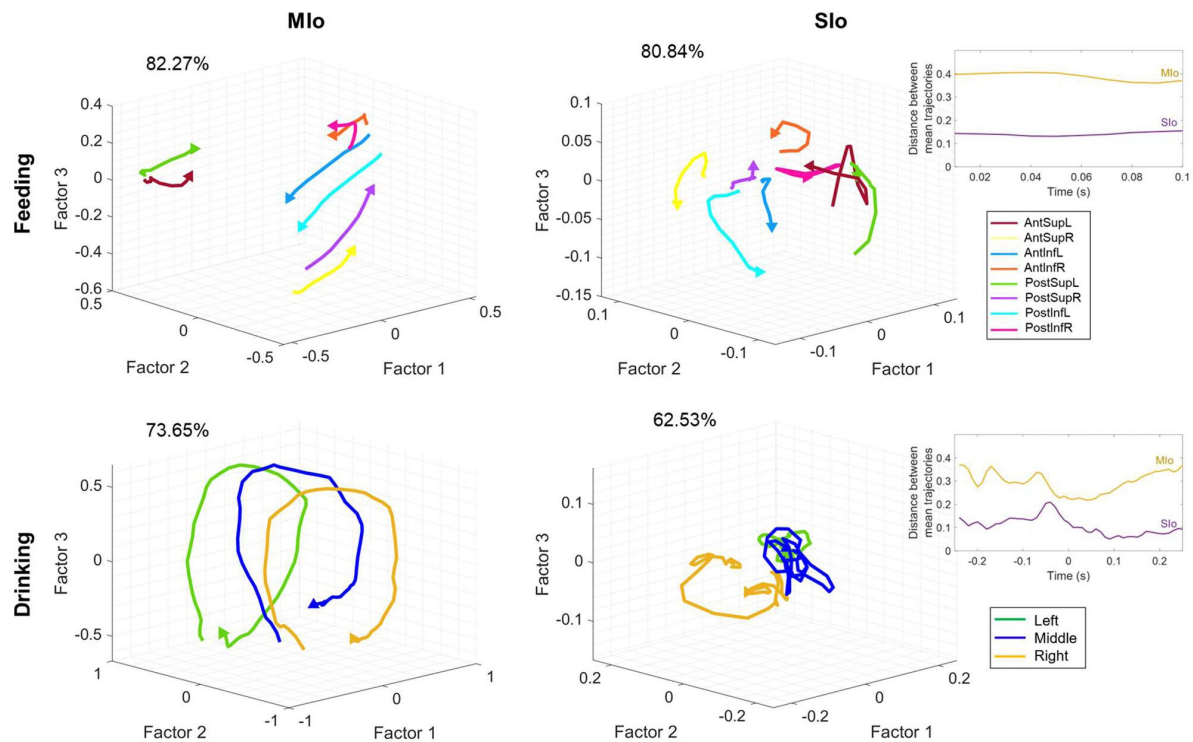


Figure 7.

Neural population trajectories vary across directions.

Trial-averaged trajectories of Mlo and Sio population activity along the first three latent factors for Monkey R, grouped by direction. Axes for Sio are 1/4 scale of Mlo. Arrows indicate the end of the trajectory. Percentages denote the sum of the variance explained by the first three factors. Inset plots show the difference between the average inter-trajectory distances of Mlo and Sio over time for both feeding and drinking.

MIO population trajectories followed a circular path and exhibited consistent patterns based on directional components. For example (**Fig. 7** [top](#)), trajectories with upward (Sup) components (AntSupL/R, PostSupL/R) rotated opposite from trajectories with downward (Inf) components (AntInfL/R, PostInfL/R). In feeding, Factors 1 and 2 captured superior-inferior and right-left directions, respectively. In drinking, MIO trajectories exhibited consistent rotational direction regardless of spout location (**Fig. 7** [bottom left](#)), while exhibiting distinct separation of trajectories for left, center, and right spout-directed tongue movements clustering at approximately -0.5, 0, and 0.5 positions along the Factor 2 axis, respectively. Indeed, inter-trajectory distances in Factor 1 were significantly higher in feeding (t-test, $p < 0.001$, mean $\pm$ 1 SD: 0.4628 ± 0.0246) than in drinking (mean $\pm$ 1 SD: 0.1286 ± 0.0610), indicating that Factor 1 resembled directional information in feeding but a condition-independent feature of population activity in drinking. The latent factors revealed a clear organizational principle: Factor 1 predominantly captured superior-inferior directional components in feeding, while Factor 2 primarily represented left-right directional components of tongue movement in both tasks.

Similar to previous findings ([Russo et al., 2018](#) [top](#)), SIO trajectories in both feeding and drinking showed stark differences from MIO as they were more tangled and exhibited less direct (i.e., sharp turns) paths (**Fig. 7** [right](#)). Unlike MIO trajectories, SIO trajectories spanned a smaller neural space, had variable distances between trajectories, and showed inconsistent patterns based on directional components. Quantitative analysis revealed greater separation between trial-averaged population trajectories in MIO compared to SIO (**Fig. 7** [inset](#), t-test, $p < 0.01$, mean $\pm$ 1 SD: MIO: 0.34 ± 0.03 ; SIO: 0.12 ± 0.02). These results were consistent with significantly longer distance travelled by MIO population trajectories compared to SIO in both tasks for Monkey R only (t-test, $p < 0.001$, mean $\pm$ 1 SD: feeding: MIO: 0.43 ± 0.07 ; SIO: 0.26 ± 0.09 ; drinking: MIO: 2.85 ± 0.42 ; SIO: 1.36 ± 0.52). The regional differences cannot be attributed to fewer SIO neurons used (Figure 7 – figure supplement 3).

Effects of nerve block

Sensation plays a key role in tongue positioning and movements for natural behaviors. During ingestion, tactile feedback is necessary for locating the bolus, preventing tongue bites, feeling where the drinking spout is, and identifying when it is safe to swallow. To evaluate the role of oral sensation, we used a bilateral oral nerve block to temporarily eliminate tactile sensation in the oral cavity and observe how the control of tongue movement was impacted. Below, we show how the loss of sensation affected both tongue kinematics and directional tuning of neurons during feeding and drinking. To verify that differences between the control and nerve block conditions were due to the loss of sensory feedback and not because of other factors such as sedation and injection, a sham experiment was conducted where saline was administered to the injection sites instead of nerve block. No significant changes to tongue kinematics were observed following the sham experiments (**Fig. 8** [top](#)).

Changes to tongue kinematics

In feeding, the mean and overall spread of directions were significantly different between the control and nerve block conditions (t-test, $p < 0.01$ and f-test, $p < 0.001$). There was a shift towards a smaller range of 3D directions in Monkey R, whereas there was a shift towards a broader distribution in Monkey Y under the nerve block condition (**Fig. 8A** [top](#)). The positions of maximum protrusion of the tongue during drinking, i.e., the endpoints, were also affected by the loss of sensation. These endpoints represent the planned target position of the tongue to receive the juice reward from a specific spout. In the control drinking task, the endpoints for each spout location were very distinct. In contrast, the endpoints of tongue movements in nerve block exhibited a greater overlap across locations and more variance in all three axes of motion, i.e., Posterior-Anterior, Inferior-Superior, and Left-Right (**Fig. 8B** [top](#)).

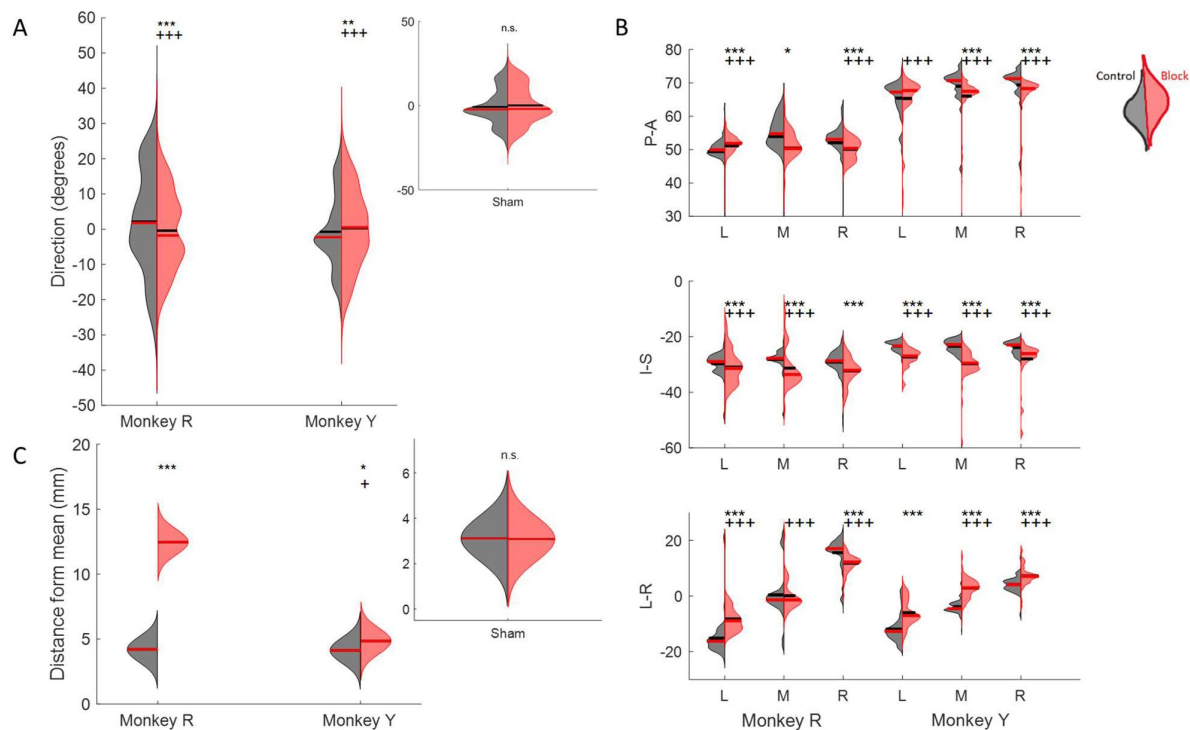


Figure 8.

Effect of nerve block on direction of tongue movement.

(A) Distribution of tongue directions during feeding. **(B)** Variance in 3D trajectory endpoints during drinking (Posterior-Anterior, Inferior-Superior, Left-Right) for each direction: left (L), middle (M), right (R). **(C)** Variation in the distance of drinking endpoint positions from the mean endpoint. Left halves of hemi-violins (black) are control and right halves (red) are nerve block for an individual. Horizontal black lines represent the mean and horizontal red lines the median. Results of two-tailed t-test and f-test are indicated by asterisks and crosses, respectively: *, † $p < 0.05$; **, †† $p < 0.01$; ***, ††† $p < 0.001$. Smaller inset plots show that there was no effect in the sham nerve block condition, for reference. The sham procedure was identical to the nerve block, except the anesthetic was substituted with saline solution

Compared to the control, the trajectories of the tongue tip in the nerve block condition during drinking had a smaller range of Left-Right values. Visually, the tongue trajectories toward the different spout locations were messier and less distinct in failed cycles where the tongue tip missed the location of the correct spout by more than ± 2 SD from the mean (**Fig. 9C**). In both monkeys there was a significant increase in the average distance from the mean endpoint position, though this difference was much greater in Monkey R (**Fig. 8C**). We noted a difference between subjects in the frequency of failed cycles and the range of left-right tongue movements under nerve block. This may reflect a possible compensatory strategy of reaching the drinking spouts with an adjacent region of the tongue, instead of contacting the right or left spout with the ipsilateral tongue in Monkey R. We observed a decrease in the average speed (t-test, Monkey R: $p < 0.001$, Monkey Y: $p > 0.1$) and an increase in the variance of the speed (F-test $p < 0.001$) of tongue movement during drinking with nerve block.

Decreased directional tuning of MIO and SIO neurons

Loss of oral sensation also affected the proportion of directionally tuned neurons and the overall distribution of PDs, though the pattern of changes differed between subjects. Following nerve block, MIO and SIO showed significant decreases in the proportion of directionally modulated neurons across both tasks (**Fig. 10A**; Chi-square, MIO: $p < 0.001$, SIO: $p < 0.05$). To confirm this effect was not merely due to altered kinematics, we conducted parallel analyses using carefully subsampled trials with matched kinematic profiles from both control and nerve-blocked conditions. This controlled analysis confirmed the persistent decrease in directional tuning during nerve block (Figure 10 – figure supplement 1).

We further investigated whether neurons gaining or losing directional selectivity differed across regions. During feeding, MIO and SIO exhibited similar proportions of neurons gaining or losing directional tuning (**Fig. 10B-top row**, Chi-square, $p > 0.1$). The drinking task revealed subject-specific differences (**Fig. 10B-bottom row**): in Monkey R, significantly more neurons lost directional tuning in SIO compared to MIO ($p < 0.01$), while in Monkey Y, SIO showed a higher proportion of neurons gaining directional tuning than MIO ($p < 0.05$). Comparing across behaviors, Monkey R demonstrated consistent patterns of directional tuning changes (Chi-square, $p > 0.1$), whereas Monkey Y showed significantly higher percentages of neurons gaining directionality during drinking than feeding in both MIO and SIO ($p < 0.05$). Interestingly, there was a large proportion (40%) of SIO neurons in Monkey Y that gained directional tuning following sensory loss compared to Monkey R (8%) during drinking.

Nerve block significantly altered PD distributions during both tasks. During feeding, MIO neurons in both subjects exhibited a significant clockwise shift in mean PD toward the center (0°), resulting in more uniform distributions (**Fig. 11A**; circular k-test, $p < 0.01$). In contrast, SIO neurons showed subject-specific responses, with only Monkey R demonstrating a significant counterclockwise shift ($p < 0.05$). During drinking under nerve block, MIO neurons displayed subject-dependent directional shifts. In Monkey R, the proportion of neurons with rightward PDs decreased and increased in all other directions whereas Monkey Y showed the opposite with increased neurons with rightward PDs (**Fig. 11B**; Chi-square, Y: $p = 0.04$). Meanwhile, SIO neurons consistently shifted rightward in both animals (Chi-square, R: $p = 0.02$), suggesting differential regional responses to peripheral deafferentation.

Separation of neural population trajectories was reduced in MIO

Disruption of tactile inputs during feeding had opposite effects on MIO and SIO. Neural population trajectories in MIO showed reduced inter-trajectory distances during nerve block compared to control conditions (**Fig. 12**, left, t-test, $p < 0.05$ in $> 89\%$ of pairs), whereas SIO exhibited increased

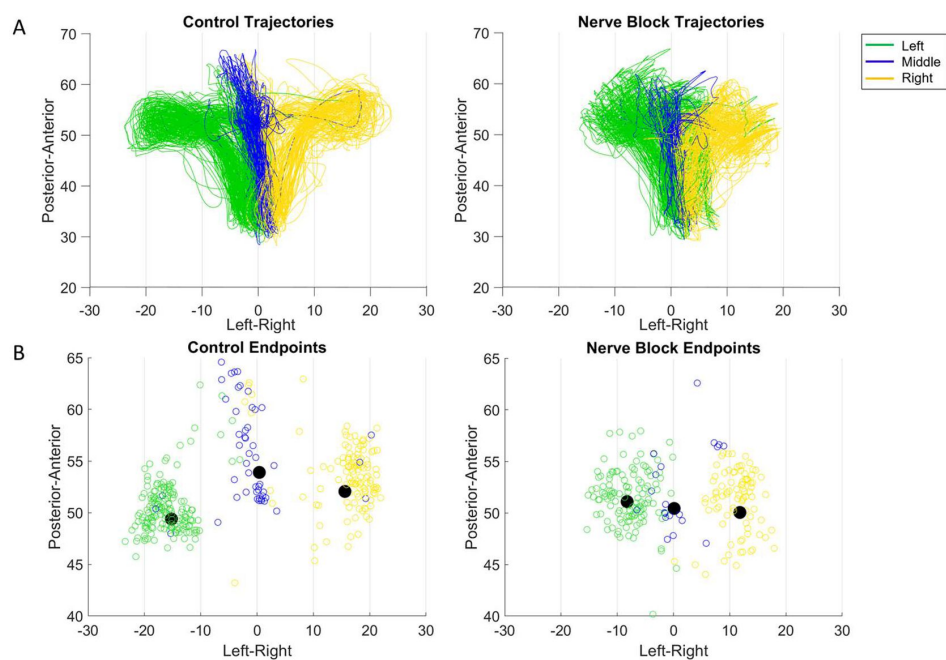


Figure 9.

Effect of nerve block on drinking kinematics in Monkey R.

- (A)** Tongue tip trajectories from starting position to one of three drinking spouts in the control and nerve block conditions.
(B) Drinking trajectory endpoints, where the black dot represents the mean endpoint position.

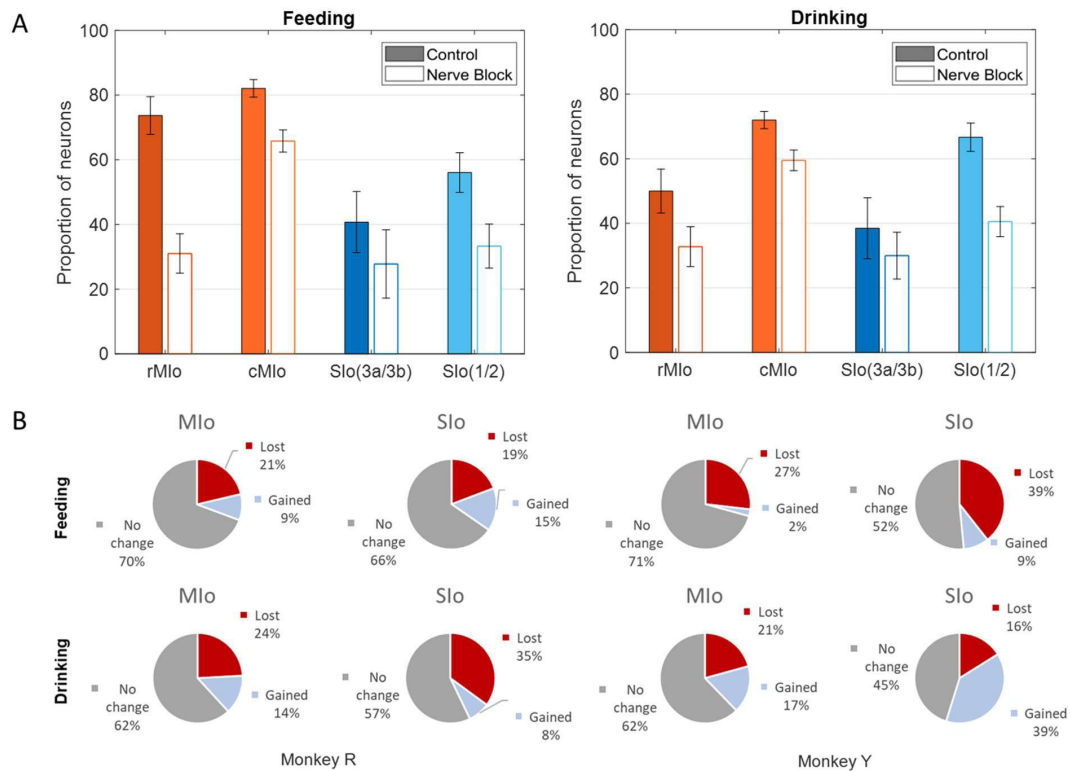


Figure 10.

Effects of nerve block on directional tuning of OSMCx neurons during feeding and drinking tasks.

(A) Percentage of directionally tuned neurons in four areas: rMlo - rostral M1, cMlo - caudal M1, Slo(3a/3b) -area 3a/3b, and Slo(1/2) -area 1/2. Filled in bars represent control while empty bars represent nerve block. Error bars represent ± 1 SE. **(B)** Percentage of Mlo and Slo neurons which gained or lost directionality with the addition of nerve block.

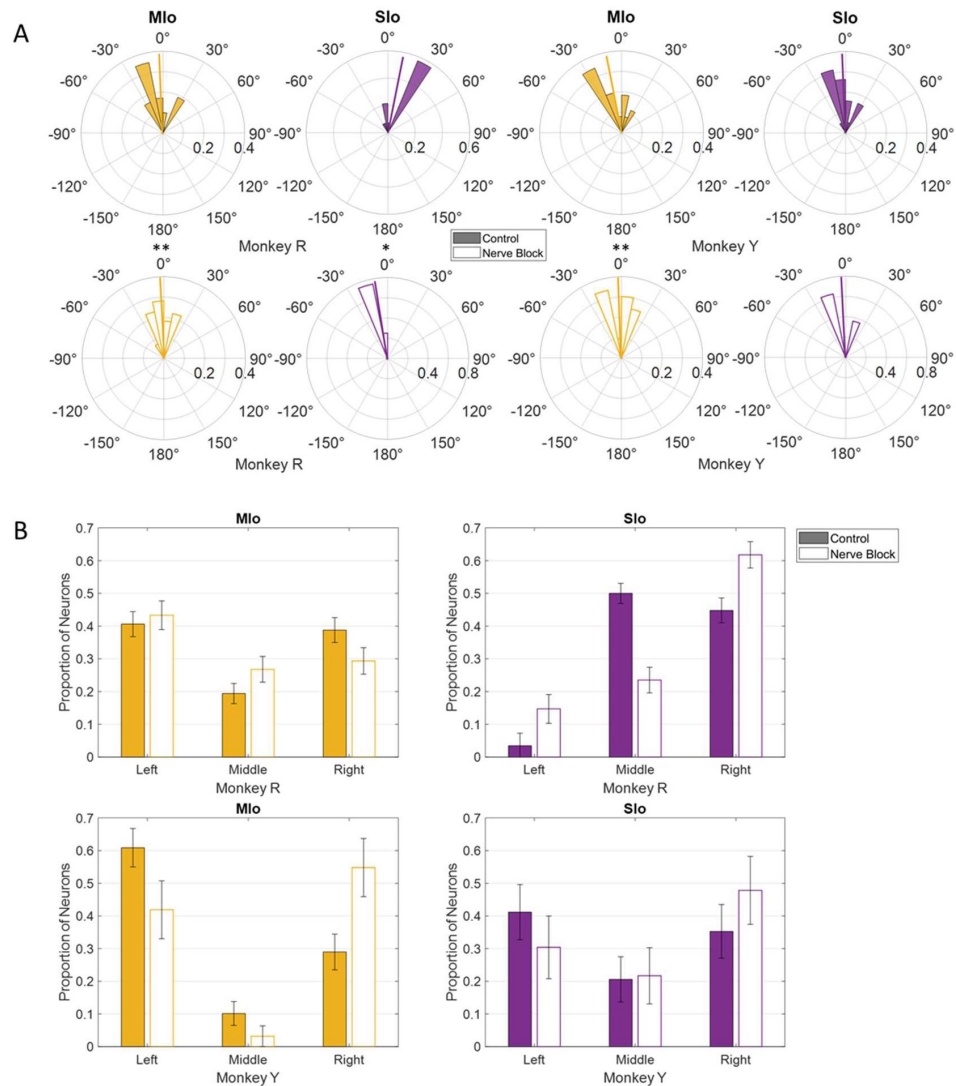


Figure 11.

Effects of nerve block on the distribution of PDs of MIO (yellow) and SIO (purple) neurons.

(A) For the feeding task, polar plots are split into 10° bins with thick colored lines representing the mean PD. Significant circular concentration test (k-test) comparing control and nerve block are indicated by asterisks: *p < 0.05; **p < 0.01; ***p < 0.001. **(B)** For the drinking task, error bars represent ± SE. Filled in bars represent control while empty bars represent nerve block.

inter-trajectory distances (**Fig. 12** top right, t-test, $p < 0.05$ in >75% of pairs). In drinking, inter-trajectory distances in both MIO and SIO were significantly reduced across all pairs (two-tailed t-test, $p < 0.01$) except middle-right in Monkey R's SIO neurons ($p > 0.1$).

The effects of nerve block on the distance travelled by population trajectories were inconsistent across subjects and behavior. Following nerve block, the total distance travelled by SIO trajectories became longer in different behaviors for a specific subject (t-test, Monkey R feeding: $p < 0.001$; Monkey Y drinking: $p < 0.05$). In contrast, MIO trajectories became shorter in drinking (Monkey Y, $p < 0.01$).

Population decoding of tongue direction

To assess directional information within population activity further, we implemented two decoding approaches to predict tongue movement direction from neuronal spiking patterns: k-nearest neighbor (KNN) classifier and long short-term memory (LSTM) neural network. Consistent with previous study on a cued tongue protrusion task (Arce et al., 2013), we found that the 3D direction of tongue movements in naturalistic behaviors could be decoded from simultaneously recorded MIO and SIO populations. The KNN classifier successfully decoded 3D tongue movement direction above chance level across behaviors and experimental conditions (**Fig. 13A**). Results of analyses using multiple linear regression model with interactions revealed several key factors affecting decoder performance: behavior type ($p < 0.001$, drinking outperformed feeding by 11%), cortical region ($p < 0.001$, MIO exceeded SIO by 13%), and inter-subject variability with behavior ($p < 0.001$, 12% higher for Monkey R during drinking, comparable accuracy during feeding). Notably, disrupting tactile sensation through nerve block did not significantly impair KNN classifier performance ($p > 0.1$). The results using LSTM were different from that of KNN; decoding tongue direction using LSTM showed substantially higher performance when using MIO neural activity (mean $R^2 \pm 1$ SD: control: $0.46-0.81 \pm 0.1$, nerve block: $0.26-0.7 \pm 0.1$) compared to SIO (mean $R^2 \pm 1$ SD: control: $0.12-0.43 \pm 0.1$, nerve block: $0.05-0.17 \pm 0.07$) across all experimental conditions (**Fig. 13B**, t-test, $p < 0.001$). This regional difference became particularly pronounced during nerve block, where SIO decoding accuracy decreased substantially more than MIO, suggesting differential reliance on tactile feedback between these cortical regions. Combining MIO and SIO showed significantly better decoder performance compared to performance using neuronal populations separately (mean $R^2 \pm 1$ SD: control: $0.78-0.92 \pm 0.05$, nerve block: $0.58-0.91 \pm 0.05$, $p < 0.001$) in feeding, but not drinking. To address the potential confound of varying population sizes between MIO and SIO, we standardized comparisons by downsampling all populations to match our smallest recorded group ($N = 28$ neurons). Decoding accuracy improved by up to 10% when using all neurons in MIO or SIO compared to using subsamples of neurons. Decoding using LSTM showed consistently higher accuracies in feeding compared to drinking regardless of the length of intervals used (100 ms, 500 ms), behavioral window (± 500 ms relative to minimum protrusion, between maximal protrusions), and directional angles (actual vs. $\{-45, 0, 45\}$). These results suggest that continuous and non-linear decoding is better suited for feeding than drinking behavior.

Discussion

Our study investigated the representation of 3D directional information in MIO and SIO during natural feeding and drinking behaviors, and how cortical representations change when tactile sensation is disrupted. By simultaneously recording large-scale cortical activity and 3D tongue kinematics, we revealed a nuanced neural encoding of tongue movement direction that varies systematically across cortical regions, behaviors, and sensory feedback conditions. We found that a substantial proportion of neurons exhibit directional tuning characterized by diverse tuning curve properties (PD, tuning shape, modulation depth). Neural population trajectories demonstrated distinct patterns across different movement directions. Directional tuning in both individual- and population-level is more robust in MIO. Following sensory loss, alterations in

Figure 12.

Effect of nerve block on population trajectories.

Trial-averaged trajectories of MIO and SIO population activity for Monkey R's feeding and drinking sessions, grouped by direction. Axes represent the latent factors from control data, with the x-axis chosen as the factor with the highest degree of separation between directions. Lighter, dotted lines represent superimposed population trajectories in the nerve block condition. Insets show the difference between the average inter-trajectory distances for control and nerve block conditions.

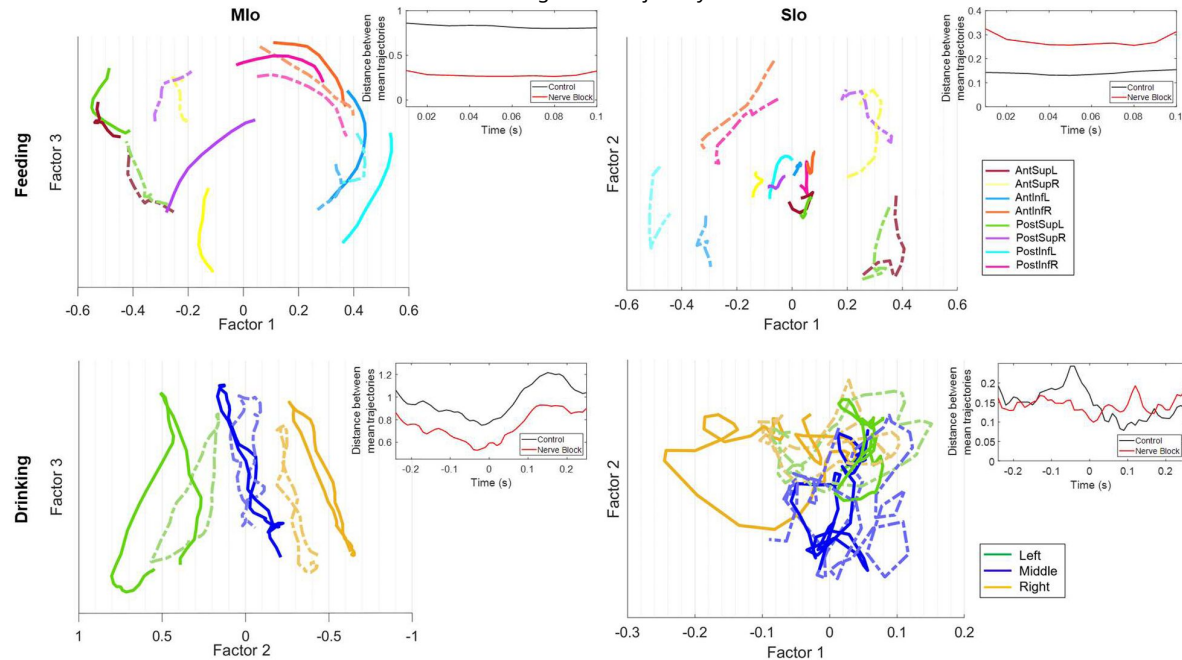
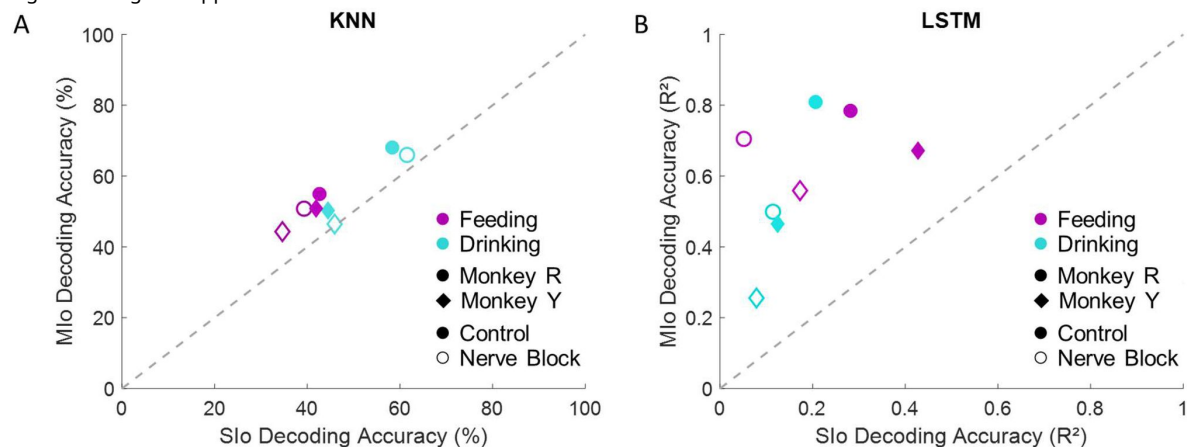


Figure 13.

Accuracies of two different decoding algorithms from MIO and SIO populations of equal size (N=28).

(A) Comparison between average decoding accuracy of KNN classifier. Chance level is 33.33%. (B) Comparison between average decoding accuracy by LSTM network. Data shown separately for each subject, behavioral task, and condition. The dashed line signifies equal decoding performance for MIO and SIO. Decoding accuracies from full populations are included in Figure 13 – figure supplement 2.



tongue kinematics were accompanied by changes in directional information in MIO and SIO, manifesting as modifications in both individual neuron tuning characteristics and the broader dynamics of population-level neural activity.

Differences across behaviors

In the present study, results from the more natural drinking are consistent with previous findings that MIO and SIO encode tongue direction during a trained protrusion task (Murray and Sessle, 1992 [↗](#); Sessle et al., 2005a [↗](#); Arce et al., 2013 [↗](#)). Our study extends this knowledge by investigating the dynamics of directional tuning of individual and population of neurons in OSMCx to 3D tongue direction during naturalistic behaviors. Unlike previous similar studies, the monkeys were not trained to reach specific targets and were instead allowed to eat and drink relatively naturally. By comparing two naturalistic behaviors, we found that the directional information in OSMCx was higher in feeding than in the drinking task, as seen in higher proportion of directionally tuned neurons, cumulative variance explained by latent factors, and decoding accuracies. That directional tuning is modifiable is consistent with previous findings in primate motor cortex where directional tuning was modulated by movement parameters such as speed, posture, distance (Aflalo and Graziano, 2006 [↗](#)) and by varying task contexts such as availability of prior information (Rickert et al., 2009 [↗](#)), individual vs. segmented arm movements (Ben-Shaul et al., 2004 [↗](#)), one-vs two-target reaching (Ebina et al., 2024 [↗](#)). A high degree of similarity in neural modes have been reported across different wrist tasks in 1-D and 2D (Gallego et al., 2018 [↗](#)). This suggests that in our study, feeding and drinking may reflect more distinct biomechanical constraints and sensorimotor requirements compared to the wrist tasks. In feeding, the tongue moves in varied directions to position the food between left-right tooththrows during chewing, whereas in the drinking task, the tongue moves to discrete and fixed spout locations. Additionally, feeding and drinking engage the jaw differently. During feeding, the jaw moves more extensively and in multiple directions, while jaw movements during drinking are smaller and primarily vertical. Lastly, the tongue-jaw coordination differs between tasks; maximum tongue protrusion coincides with maximum gape in drinking but with minimum gape in the feeding behavior (Laurence-Chasen et al., 2022 [↗](#); Punacha et al., 2024 [↗](#)). Indeed, MIO trajectories' predominant latent factor contained directional information in the feeding task, resembling superior-inferior directional components which rotated in opposite directions. In contrast, MIO trajectories in drinking exhibited a consistent rotational direction regardless of spout location (Fig. 7 [↗](#)). This may reflect a predominant non-directional information such as condition-independent time-varying spiking activity during drinking (Kaufman et al., 2013 [↗](#); Kobak et al., 2016 [↗](#); Arce-McShane et al., 2023 [↗](#)).

Comparison between MIO and SIO

Directional tuning of neurons during feeding showed a notable disparity between MIO and SIO, suggesting that MIO carries more robust directional information for tongue movements in feeding tasks. The similar proportions of directionally tuned MIO and SIO neurons in the drinking task studied here were consistent with previous findings (Arce et al., 2013 [↗](#)). At the population level, MIO trajectories showed more consistent rotational patterns and greater inter-trajectory separation than those in SIO. Consistent with results from previous studies (Michaels et al., 2016 [↗](#); Seely et al., 2016 [↗](#); Russo et al., 2018 [↗](#)), MIO trajectories exhibited low tangling and smoother dynamics than SIO trajectories. These may suggest that the low tangling in MIO confers noise robustness while the higher tangling in SIO reflects variability in the tactile signals received by SIO during feeding. Consistent with our previous study (Laurence-Chasen et al., 2023 [↗](#)), decoding from MIO yielded higher accuracies than from SIO in both behaviors. These results support the well-established role of MIO in the control of movement parameters, especially direction. Varying tongue shape, tongue's contact with varying bolus properties (size and texture) and other oral structures (palate, teeth) may weaken the directional signal contained in SIO activity. Thus, small differences in tongue kinematics might create large differences in sensory signals across trials. When looking at trial-averaged signals, this natural variability could make the neural response

patterns appear less precise or specific than they are. These are consistent with our findings that for both tasks, spiking variability was higher in SIO, and that the variance accounted for was lower in SIO population activity compared to MIO.

Laterality in OSMCx

Similar to previous results in arm motor cortex (Lillicrap and Scott, 2013 [↗](#)), we observed non-uniform PD distributions consistent with the frequency distribution of tongue movements, suggesting that neural populations contain information that reflects the anatomical constraints of the tongue. The highest frequency of both observed directions and directional tuning peaks were in the anterior and superior directions. We additionally found that the peak of the PD distribution, especially in feeding, coincides with leftward tongue movements, suggesting the presence of laterality in the PDs of OSMCx neurons. Previous results in humans examined using fMRI reported that hemispheric differences in sensorimotor activity during voluntary tongue movements are related to the preferred chew side (Shinagawa et al., 2003 [↗](#)). This was not the case in our study as the preferred chew side was the same for both monkeys (i.e., right side), despite differences in the predominant PD. It is possible that the difference between the two subjects is related to the difference in recording locations, with Monkey Y's being more lateral and therefore closer to the swallow area of the cortex than Monkey R's (Supplementary file 1, **Fig. 1** [↗](#)). Monkey Y had a higher proportion of neurons that were tuned to tongue direction during feeding compared to Monkey R (Figure 3 – figure supplement 2), but fewer during drinking. An avenue for further study could be a unilateral nerve block on the preferred side to measure how the unaffected side of the tongue compensates for the lack of sensation in the affected side. A previous study found that unilateral lingual nerve transection in pigs alters the coordination of the ipsilateral tongue side during chewing (Montuelle et al., 2020 [↗](#)). The tongue is a complex group of muscles, with intrinsic muscles primarily contributing to the shape of the tongue and extrinsic muscles contributing more to the positioning of the tongue. Therefore, it is possible that the neurons which are strongly tuned to tongue direction have direct connections to the extrinsic muscles on the ipsilateral side. Looking at how each side of the tongue responds independently to unilateral nerve block, and how this interacts with directional preference may give us more information about how the unique structure of the tongue is coordinated.

Role of tactile feedback

Previously, we reported that the administration of bilateral nerve block to the sensory branches of the trigeminal nerve impaired feeding performance and tongue jaw coordination (Laurence-Chasen et al., 2022 [↗](#)). The present study extends these findings by showing that directional movement of the tongue (*kinematics*) and the directional information in both MIO and SIO are also affected by the loss of tactile inputs from the tongue and other structures of the oral cavity (e.g., palate, teeth, gingiva). These findings highlight the critical role of sensory information in sensorimotor control in general (Dadarlat et al., 2015 [↗](#); Delhay et al., 2018 [↗](#)) and in the representation and computation of directional signals for controlling tongue movements. MIO and SIO neurons, which respond to tactile and proprioceptive inputs from the tongue (Huang et al., 1989 [↗](#); Lin et al., 1994 [↗](#); Toda and Taoka, 2002 [↗](#), 2004 [↗](#), 2006 [↗](#); Arce et al., 2013 [↗](#); Cerkevich et al., 2014 [↗](#)), use sensory information to plan and adjusts tongue movements to achieve contact with the spout or position the bolus appropriately at different stages of the feeding sequence. Without tactile feedback, subjects may rely on alternative sensory cues like taste for locating the spout or bolus (Todorank and Bartoshuk, 1991 [↗](#)). In a recent optogenetic inhibition study on licking in mice, it was found that the tongue/jaw regions of the somatosensory cortex were necessary for proper tongue targeting but not for the core motor capabilities of the tongue (Xu et al., 2022 [↗](#)). The reduced range of tongue motion we observed likely stems from sensory loss rather than motor impairment. While our experimental setup did not eliminate visual feedback that monkeys might use to readjust tongue position in the drinking task, oral sensory loss alone had a significant effect on the monkeys' performance in feeding, as the tongue was out of sight within the oral cavity.

Individual differences were notable in our study possibly due to differences in the electrode array placement or compensatory strategies. This individual variability suggests that studying additional subjects would provide valuable insights into how OSMCx adapts following sensory disruption. In our study, head position and hand movement were restrained, to eliminate contributions of hand-to-mouth movements when handling food or drink. The hand and orofacial cortical areas are anatomically adjacent and highly interconnected (Forrester and Rodriguez, 2015 [↗](#)), and researchers have found a neural region in mice that coordinates hand-to-mouth movements during natural feeding (An et al., 2022 [↗](#)). Truly natural feeding would involve holding food up to the mouth, as well as free head movement, which would make tracking of the marker positions difficult under this experimental setup. Advances in tracking tongue movements would be necessary to study more complex feeding sequences.

Clinical implications

This study offers new information about the important role of sensorimotor integration in controlling tongue direction during natural behaviors. There is a high degree of directional information contained in the spiking activity of the orofacial cortex, especially in the motor areas. The effect of the bilateral nerve block serves to enhance our understanding of the processes affected by oral sensorimotor dysfunctions such as trigeminal neuropathies. It demonstrates the importance of oral sensation for supporting the full range of directional motion but also shows that significant directional information can be extracted even in the absence of tactile feedback. This type of knowledge can inform the diagnosis and rehabilitation of orolingual dysfunctions, following stroke or glossectomy. There have also been advancements in brain-computer interface (BCI) by decoding the real-time signals of arm region of the motor cortex to control prosthetic arm movement (Collinger et al., 2013 [↗](#); Vilela and Hochberg, 2020 [↗](#)) or muscle stimulation (Ethier and Miller, 2015 [↗](#)), as well as efforts to restore sensory feedback by stimulating correct areas of somatosensory cortex in response to sensors on a prosthetic (Tabot et al., 2013 [↗](#); Flesher et al., 2021 [↗](#)). That the OSMCx, particularly MIO, can rapidly decode tongue direction during natural behaviors is significant for developing neuroprosthetic control or soft prosthetics.

Additional information

Ethics statement

Experiments were performed in the University of Chicago XROMM Facility. All protocols were approved by the University of Chicago Animal Care and Use Committee and complied with the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

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Author contributions

Victoria B. Hosack: data analysis, writing – original draft, writing – review and editing, visualization.

Fritzie I. Arce-McShane: conceptualization, methodology, investigation, writing – review and editing, supervision, project administration, funding acquisition.

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Additional files

Supplementary file 1 [🔗](#)

Supplementary file 2 [🔗](#)

Additional information

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Reviewer #1 (Public review):

Summary:

Hosack and Arce-McShane investigate how the 3D movement direction of the tongue is represented in the orofacial part of the sensory-motor cortex and how this representation changes with the loss of oral sensation. They examine the firing patterns of neurons in the orofacial parts of the primary motor cortex (MIo) and somatosensory cortex (SIo) in non-human primates (NHPs) during drinking and feeding tasks. While recording neural activity, they also tracked the kinematics of tongue movement using biplanar video-radiography of

markers implanted in the tongue. Their findings indicate that many units in both MIO and SIO are directionally tuned during the drinking task. However, during the feeding task, directional turning was more frequent in MIO units and less prominent in SIO units. Additionally, in some recording sessions, they blocked sensory feedback using bilateral nerve block injections, which seemed to result in fewer directionally tuned units and changes in the overall distribution of the preferred direction of the units.

Strengths:

The most significant strength of this paper lies in its unique combination of experimental tools. The author utilized a video-radiography method to capture 3D kinematics of the tongue movement during two behavioral tasks while simultaneously recording activity from two brain areas. This specific dataset and experimental setup hold great potential for future research on the understudied orofacial segment of the sensory-motor area.

Weaknesses:

A substantial portion of the paper is dedicated to establishing directional tuning in individual neurons, followed by an analysis of how this tuning changes when sensory feedback is blocked. While such characterizations are valuable, particularly in less-studied motor cortical areas and behaviors, the discrepancies in tuning changes across the two NHPs, coupled with the overall exploratory nature of the study, render the interpretation of these subtle differences somewhat speculative. At the population level, both decoding analyses and state space trajectories from factor analysis indicate that movement direction (or spout location) is robustly represented. However, as with the single-cell findings, the nuanced differences in neural trajectories across reach directions and between baseline and sensory-block conditions remain largely descriptive. To move beyond this, model-based or hypothesis-driven approaches are needed to uncover mechanistic links between neural state space dynamics and behavior.

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Reviewer #2 (Public review):

Summary:

This manuscript by Hosack and Arce-McShane examines the directional tuning of neurons in macaque primary motor (MIO) and somatosensory (SIO) cortex. The neural basis of tongue control is far less studied than, for example, forelimb movements, partly because the tongue's kinematics and kinetics are difficult to measure. A major technical advantage of this study is using biplanar video-radiography, processed with modern motion tracking analysis software, to track the movement of the tongue inside the oral cavity. Compared to prior work, the behaviors are more naturalistic behaviors (feeding and licking water from one of three spouts), although the animals were still head-fixed.

The study's main findings are that:

- A majority of neurons in MIO and a (somewhat smaller) percentage of SIO modulated their firing rates during tongue movements, with different modulation depending on the direction of movement (i.e., exhibited directional tuning). Examining the statistics of tuning across neurons, there was anisotropy (e.g., more neurons preferring anterior movement) and a lateral bias in which tongue direction neurons preferred that was consistent with the innervation patterns of tongue control muscles (although with some inconsistency between monkeys).
- Consistent with this encoding, tongue position could be decoded with moderate accuracy even from small ensembles of ~28 neurons.

- There were differences observed in the proportion and extent of directional tuning between the feeding and licking behaviors, with stronger tuning overall during licking. This potentially suggests behavioral context-dependent encoding.
- The authors then went one step further and used a bilateral nerve block to the sensory inputs (trigeminal nerve) from the tongue. This impaired the precision of tongue movements and resulted in an apparent reduction and change in neural tuning in Mio and SIO.

Strengths:

The data are difficult to obtain and appear to have been rigorously measured, and provide a valuable contribution to this under-explored subfield of sensorimotor neuroscience. The analyses adopt well-established methods especially from the arm motor control literature, and represent a natural starting point for characterizing tongue 3D direction tuning.

Weaknesses:

There are alternative explanations from some of the interpretations, but those interpretations are described in a way that clearly distinguishes results from interpretations, and readers can make their own assessments. Some of these limitations are described in more detail below.

One weakness of the current study is that there is substantial variability in results between monkeys.

This study focuses on describing directional tuning using the preferred direction (PD) / cosine tuning model popularized by Georgopoulos and colleagues for understanding neural control of arm reaching in the 1980s. This is a reasonable starting point and a decent first order description of neural tuning. However, the arm motor control field has moved far past that viewpoint, and in some ways an over-fixation on static representational encoding models and PDs held that field back for many years. The manuscript benefit from drawing the readers' attention (perhaps in their Discussion) that PDs are a very simple starting point for characterizing how cortical activity relates to kinematics, but that there is likely much richer population-level dynamical structure and that a more mechanistic, control-focused analytical framework may be fruitful. A good review of this evolution in the arm field can be found in Vyas S, Golub MD, Sussillo D, Shenoy K. 2020. Computation Through Neural Population Dynamics. *Annual Review of Neuroscience*. 43(1):249-75. A revised version of the manuscript incorporates more population-level analyses, but with inconsistent use of quantifications/statistics and without sufficient contextualization of what the reader is to make of these results.

The described changes in tuning after nerve block could also be explained by changes in kinematics between these conditions, which temper the interpretation of these interesting results.

I am not convinced of the claim that tongue directional encoding fundamentally changes between drinking and feeding given the dramatically different kinematics and the involvement of other body parts like the jaw (e.g., the reference to Laurence-Chasen et al. 2023 just shows that there is tongue information independent of jaw kinematics, not that jaw movements don't affect these neurons' activities). I also find the nerve block results inconsistent (more tuning in one monkey, less in the other?) and difficult to really learn something fundamental from, besides that neural activity and behavior both change - in various ways - after nerve block (not at all surprising but still good to see measurements of).

The manuscript states that "Our results suggest that the somatosensory cortex may be less involved than the motor areas during feeding, possibly because it is a more ingrained and stereotyped behavior as opposed to tongue protrusion or drinking tasks". An alternative explanation be more statistical/technical in nature: that during feeding, there will be more

variability in exactly what somatosensation afferent signals are being received from trial to trial (because slight differences in kinematics can have large differences in exactly where the tongue is and the where/when/how of what parts of it are touching other parts of the oral cavity)? This variability could "smear out" the apparent tuning using these types of trial-averaged analyses. Given how important proprioception and somatosensation are for not biting the tongue or choking, the speculation that somatosensory cortical activity is suppressed during feedback is very counter-intuitive to this reviewer. In the revised manuscript the authors note these potential confounds and other limitations in the Discussion.

<https://doi.org/10.7554/eLife.101325.2.sa2>

Reviewer #3 (Public review):

Summary

In this study, the authors aim to uncover how 3D tongue direction is represented in the Motor (M1o) and Somatosensory (S1o) cortex. In non-human primates implanted with chronic electrode arrays, they use X-ray based imaging to track the kinematics of the tongue and jaw as the animal is either chewing food or licking from a spout. They then correlate the tongue kinematics with the recorded neural activity. They perform both single-unit and population level analyses during feeding and licking. Then, they recharacterize the tuning properties after bilateral lidocaine injections in the two sensory branches of the trigeminal nerve. They report that their nerve block causes a reorganization of the tuning properties and population trajectories. Overall, this paper concludes that M1o and S1o both contain representations of the tongue direction, but their numbers, their tuning properties and susceptibility to perturbed sensory input are different.

Strengths

The major strengths of this paper are in the state-of-the-art experimental methods employed to collect the electrophysiological and kinematic data. In the revision, the single-unit analyses of tuning direction are robustly characterized. The differences in neural correlations across behaviors, regions and perturbations are robust. In addition to the substantial amount of largely descriptive analyses, this paper makes two convincing arguments 1) The single-neuron correlates for feeding and licking in OSMCx are different - and can't be simply explained by different kinematics and 2) Blocking sensory input alters the neural processing during orofacial behaviors. The evidence for these claims is solid.

Weaknesses

The main weakness of this paper is in providing an account for these differences to get some insight into neural mechanisms. For example, while the authors show changes in neural tuning and different 'neural trajectory' shapes during feeding and drinking - their analyses of these differences are descriptive and provide limited insight for the underlying neural computations.

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Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

In their paper, Hosack and Arce-McShane investigate how the 3D movement direction of the tongue is represented in the orofacial part of the sensory-motor cortex and how this representation changes with the loss of oral sensation. They examine the firing patterns of neurons in the orofacial parts of the primary motor cortex (MIO) and somatosensory cortex (SIO) in non-human primates (NHPs) during drinking and feeding tasks. While recording neural activity, they also tracked the kinematics of tongue movement using biplanar videoradiography of markers implanted in the tongue. Their findings indicate that most units in both MIO and SIO are directionally tuned during the drinking task. However, during the feeding task, directional turning was more frequent in MIO units and less prominent in SIO units. Additionally, in some recording sessions, they blocked sensory feedback using bilateral nerve block injections, which resulted in fewer directionally tuned units and changes in the overall distribution of the preferred direction of the units.

Strengths:

The most significant strength of this paper lies in its unique combination of experimental tools. The author utilized a video-radiography method to capture 3D kinematics of the tongue movement during two behavioral tasks while simultaneously recording activity from two brain areas. Moreover, they employed a nerve-blocking procedure to halt sensory feedback. This specific dataset and experimental setup hold great potential for future research on the understudied orofacial segment of the sensory-motor area.

Weaknesses:

Aside from the last part of the result section, the majority of the analyses in this paper are focused on single units. I understand the need to characterize the number of single units that directly code for external variables like movement direction, especially for less-studied areas like the orofacial part of the sensory-motor cortex. However, as a field, our decadelong experience in the arm region of sensory-motor cortices suggests that many of the idiosyncratic behaviors of single units can be better understood when the neural activity is studied at the level of the state space of the population. By doing so, for the arm region, we were able to explain why units have "mixed selectivity" for external variables, why the tuning of units changes in the planning and execution phase of the movement, why activity in the planning phase does not lead to undesired muscle activity, etc. See (Gallego et al. 2017; Vyas et al. 2020; Churchland and Shenoy 2024) for a review. Therefore, I believe investigating the dynamics of the population activity in orofacial regions can similarly help the reader go beyond the peculiarities of single units and in a broader view, inform us if the same principles found in the arm region can be generalized to other segments of sensorymotor cortex.

We thank and agree with the reviewer on the value of information gained from studying population activity. We also appreciate that population analyses have led to the understanding that individual neurons have "mixed selectivity". We have shown previously that OSMCx neurons exhibit mixed selectivity in their population activity and clear separation between latent factors associated with gape and bite force levels (Arce-McShane FI, Sessle BJ, Ram Y, Ross CF, Hatsopoulos NG (2023) Multiple regions of primate orofacial sensorimotor cortex encode bite force and gape. *Front Systems Neurosci.* doi: 10.3389/fnsys.2023.1213279. PMID: 37808467 PMCID: 10556252), and chew-side and food types (Li Z & Arce-McShane FI (2023). Cortical representation of mastication in the primate

orofacial sensorimotor cortex. Program No. NANO06.05. 2023 Neuroscience Meeting Planner. Washington, D.C.: Society for Neuroscience, 2023. Online.).

The primary goal of this paper was to characterize single units in the orofacial region and to do a follow-up paper on population activity. In the revised manuscript, we have now incorporated the results of population-level analyses. The combined results of the single unit and population analyses provide a deeper understanding of the cortical representation of 3D direction of tongue movements during natural feeding and drinking behaviors.

Further, for the nerve-blocking experiments, the authors demonstrate that the lack of sensory feedback severely alters how the movement is executed at the level of behavior and neural activity. However, I had a hard time interpreting these results since any change in neural activity after blocking the orofacial nerves could be due to either the lack of the sensory signal or, as the authors suggest, due to the NHPs executing a different movement to compensate for the lack of sensory information or the combination of both of these factors. Hence, it would be helpful to know if the authors have any hint in the data that can tease apart these factors. For example, analyzing a subset of nerve-blocked trials that have similar kinematics to the control.

Thank you for bringing this important point. We agree with the reviewer that any change in the neural activity may be attributed to lack of sensory signal or to compensatory changes or a combination of these factors. To tease apart these factors, we sampled an equal number of trials with similar kinematics for both control and nerve block feeding sessions. We added clarifying description of this approach in the Results section of the revised manuscript: “To confirm this effect was not merely due to altered kinematics, we conducted parallel analyses using carefully subsampled trials with matched kinematic profiles from both control and nerve-blocked conditions.”

Furthermore, we ran additional analysis for the drinking datasets by subsampling a similar distribution of drinking movements from each condition. We compared the neural data from an equal number of trials with a similar left-right angle of movement in the last 100 ms of the tongue trajectory, nearest the spout. We compared the directional tuning across an equal number of trials with a similar left-right angle of movement in the last 100 ms of the tongue trajectory, nearest the spout. These analyses that control for similar kinematics showed that there was still a decrease in the proportion of directionally modulated neurons with nerve block compared to the control. This confirms that the results may be attributed to the lack of tactile information. These are now integrated in the revised paper under Methods section: Directional tuning of single neurons, as well as Results section: Effects of nerve block: Decreased directional tuning of MIO and SIO neurons and Figure 10 – figure supplement 1.

Reviewer #2 (Public review):

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- A majority of neurons in MIO and a (somewhat smaller) percentage of SIO modulated their firing rates during tongue movements, with different modulations depending on the direction of movement (i.e., exhibited directional tuning). Examining the statistics of tuning across neurons, there was anisotropy (e.g., more neurons preferring anterior movement) and a lateral bias in which tongue direction neurons preferred that was consistent with the innervation patterns of tongue control muscles (although with some inconsistency between monkeys).
- Consistent with this encoding, tongue position could be decoded with moderate accuracy even from small ensembles of ~28 neurons.
- There were differences observed in the proportion and extent of directional tuning between the feeding and licking behaviors, with stronger tuning overall during licking. This potentially suggests behavioral context-dependent encoding.
- The authors then went one step further and used a bilateral nerve block to the sensory inputs (trigeminal nerve) from the tongue. This impaired the precision of tongue movements and resulted in an apparent reduction and change in neural tuning in MIO and SIO.

Strengths:

The data are difficult to obtain and appear to have been rigorously measured, and provide a valuable contribution to this under-explored subfield of sensorimotor neuroscience. The analyses adopt well-established methods, especially from the arm motor control literature, and represent a natural starting point for characterizing tongue 3D direction tuning.

Weaknesses:

There are alternative explanations for some of the interpretations, but those interpretations are described in a way that clearly distinguishes results from interpretations, and readers can make their own assessments. Some of these limitations are described in more detail below.

One weakness of the current study is that there is substantial variability in results between monkeys, and that only one session of data per monkey/condition is analyzed (8 sessions total). This raises the concern that the results could be idiosyncratic. The Methods mention that other datasets were collected, but not analyzed because the imaging pre-processing is very labor-intensive. While I recognize that time is precious, I do think in this case the manuscript would be substantially strengthened by showing that the results are similar on other sessions.

We acknowledge the reviewer's concern about inter-subject variability. Animal feeding and drinking behaviors are quite stable across sessions, thus, we do not think that additional sessions will address the concern that the results could be idiosyncratic. Each of the eight datasets analyzed here have sufficient neural and kinematic data to capture neural and behavioral patterns. Nevertheless, we performed some of the analyses on a second feeding dataset from Monkey R. The results from analyses on a subset of this data were consistent across datasets; for example, (1) similar proportions of directionally tuned neurons, (2) similar distances between population trajectories (t-test $p > 0.9$), and (3) a consistently smaller distance between Anterior-Posterior pairs than others in MIO (t-test $p < 0.05$) but not SIO ($p > 0.1$).

This study focuses on describing directional tuning using the preferred direction (PD) / cosine tuning model popularized by Georgopoulos and colleagues for understanding

neural control of arm reaching in the 1980s. This is a reasonable starting point and a decent first-order description of neural tuning. However, the arm motor control field has moved far past that viewpoint, and in some ways, an over-fixation on static representational encoding models and PDs held that field back for many years. The manuscript benefits from drawing the readers' attention (perhaps in their Discussion) that PDs are a very simple starting point for characterizing how cortical activity relates to kinematics, but that there is likely much richer population-level dynamical structure and that a more mechanistic, control-focused analytical framework may be fruitful. A good review of this evolution in the arm field can be found in Vyas S, Golub MD, Sussillo D, Shenoy K. 2020. Computation Through Neural Population Dynamics. Annual Review of Neuroscience. 43(1):249-75

Thank you for highlighting this important point. Research on orofacial movements hasn't progressed at the same pace as limb movement studies. Our manuscript focused specifically on characterizing the 3D directional tuning properties of individual neurons in the orofacial area—an analysis that has not been conducted previously for orofacial sensorimotor control. While we initially prioritized this individual neuron analysis, we recognize the value of broader population-level insights.

Based on your helpful feedback, we have incorporated additional population analyses to provide a more comprehensive picture of orofacial sensorimotor control and expanded our discussion section. We appreciate your expertise in pushing our work to be more thorough and aligned with current neuroscience approaches.

Can the authors explain (or at least speculate) why there was such a large difference in behavioral effect due to nerve block between the two monkeys (Figure 7)?

We acknowledge this as a variable inherent to this type of experimentation. Previous studies have found large kinematic variation in the effect of oral nerve block as well as in the following compensatory strategies between subjects. Each animal's biology and response to perturbation vary naturally. Indeed, our subjects exhibited different feeding behavior even in the absence of nerve block perturbation (see Figure 2 in Laurence-Chasen et al., 2022). This is why each individual serves as its own control.

Do the analyses showing a decrease in tuning after nerve block take into account the changes (and sometimes reduction in variability) of the kinematics between these conditions? In other words, if you subsampled trials to have similar distributions of kinematics between Control and Block conditions, does the effect hold true? The extreme scenario to illustrate my concern is that if Block conditions resulted in all identical movements (which of course they don't), the tuning analysis would find no tuned neurons. The lack of change in decoding accuracy is another yellow flag that there may be a methodological explanation for the decreased tuning result.

Thank you for bringing up this point. We accounted for the changes in the variability of the kinematics between the control and nerve block conditions in the feeding dataset where we sampled an equal number of trials with similar kinematics for both control and nerve block. However, we did not control for similar kinematics in the drinking task. In the revised manuscript, we have clarified this and performed similar analysis for the drinking task. We sampled a similar distribution of drinking movements from each condition. We compared the neural data from an equal number of trials with a similar left-right angle of movement in the last 100 ms of the tongue trajectory, nearest the spout. There was a decrease in the percentage of neurons that were directionally modulated (between 30 and 80%) with nerve block compared to the control. These results have been included in the revised paper under

Methods section: Directional tuning of single neurons, as well as Results section: Effects of nerve block: Decreased directionality of MIO and SIO neurons.

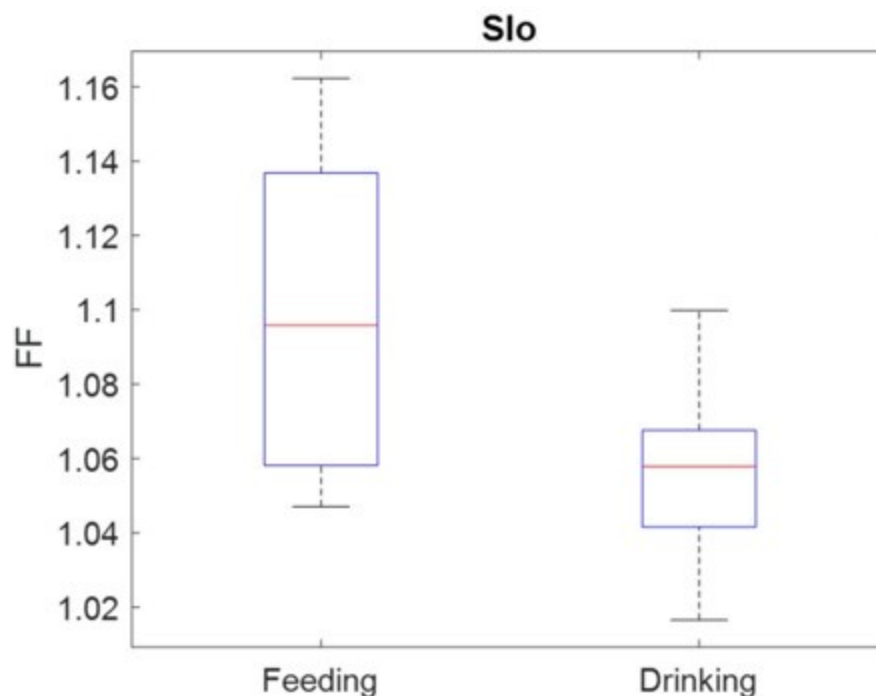
While the results from decoding using KNN did not show significant differences between decoding accuracies in control vs. nerve block conditions, the results from the additional factor analysis and decoding using LSTM were consistent with the decrease in directional tuning at the level of individual neurons.

The manuscript states that "Our results suggest that the somatosensory cortex may be less involved than the motor areas during feeding, possibly because it is a more ingrained and stereotyped behavior as opposed to tongue protrusion or drinking tasks". Could an alternative explanation be more statistical/technical in nature: that during feeding, there will be more variability in exactly what somatosensory afferent signals are being received from trial to trial (because slight differences in kinematics can have large differences in exactly where the tongue is and the where/when/how of what parts of it are touching other parts of the oral cavity)? This variability could "smear out" the apparent tuning using these types of trial-averaged analyses. Given how important proprioception and somatosensation are for not biting the tongue or choking, the speculation that somatosensory cortical activity is suppressed during feedback is very counter-intuitive to this reviewer.

Thank you for bringing up this point. We have now incorporated this in our revised Discussion (see Comparison between MIO and SIO). We agree with the reviewer that trial-by-trial variability in the afferent signals may account for the lower directional signal in SIO during feeding than in drinking. Indeed, SIO's mean-matched Fano factor in feeding was significantly higher than those in drinking (Author response image 1). Moreover, the results of the additional population and decoding analyses also support this.

Author response image 1.

Comparison of mean-matched Fano Factor between SIO neurons during feeding and drinking control tasks across both subjects (Wilcoxon rank sum test, $p < 0.001$).



Reviewer #3 (Public review):

Summary:

In this study, the authors aim to uncover how 3D tongue direction is represented in the Motor (M1o) and Somatosensory (S1o) cortex. In non-human primates implanted with chronic electrode arrays, they use X-ray-based imaging to track the kinematics of the tongue and jaw as the animal is either chewing food or licking from a spout. They then correlate the tongue kinematics with the recorded neural activity. Using linear regressions, they characterize the tuning properties and distributions of the recorded population during feeding and licking. Then, they recharacterize the tuning properties after bilateral lidocaine injections in the two sensory branches of the trigeminal nerve. They report that their nerve block causes a reorganization of the tuning properties. Overall, this paper concludes that M1o and S1o both contain representations of the tongue direction, but their numbers, their tuning properties, and susceptibility to perturbed sensory input are different.

Strengths:

The major strengths of this paper are in the state-of-the-art experimental methods employed to collect the electrophysiological and kinematic data.

Weaknesses:

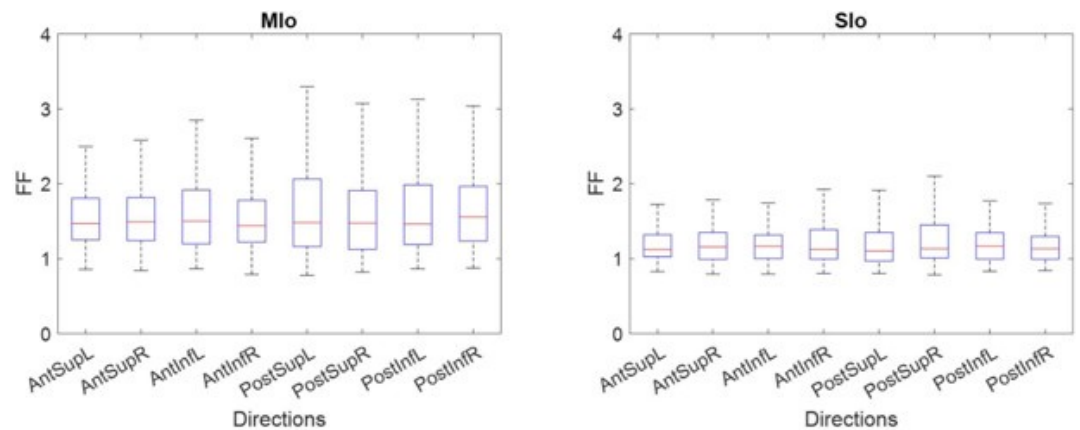
However, this paper has a number of weaknesses in the analysis of this data.

It is unclear how reliable the neural responses are to the stimuli. The trial-by-trial variability of the neural firing rates is not reported. Thus, it is unclear if the methods used for establishing that a neuron is modulated and tuned to a direction are susceptible to spurious correlations. The authors do not use shuffling or bootstrapping tests to determine the robustness of their fits or determining the 'preferred direction' of the neurons. This weakness colors the rest of the paper.

Thank you for raising these points. We have performed the following additional analyses: (1) We have added analyses to ensure that the results could not be explained by neural variability. To show the trial-by-trial variability of the neural firing rates, we have calculated the Fano factor (mean overall = 1.34747; control = 1.46471; nerve block = 1.23023). The distribution was similar across directions, suggesting that responses of M1o and S1o neurons to varying 3D directions were reliable. (2) We have used a bootstrap procedure to ensure that directional tuning cannot be explained by mere chance. (3) To test the robustness of our PDs we also performed a bootstrap test, which yielded the same results for >90% of neurons, and a multiple linear regression test for fit to a cosine-tuning function. In the revised manuscript, the Methods and Results sections have been updated to include these analyses.

Author response image 2.

Comparison of Fano Factor across directions for M1o and S1o Feeding Control (Kruskal-Wallis, $p > 0.7$).



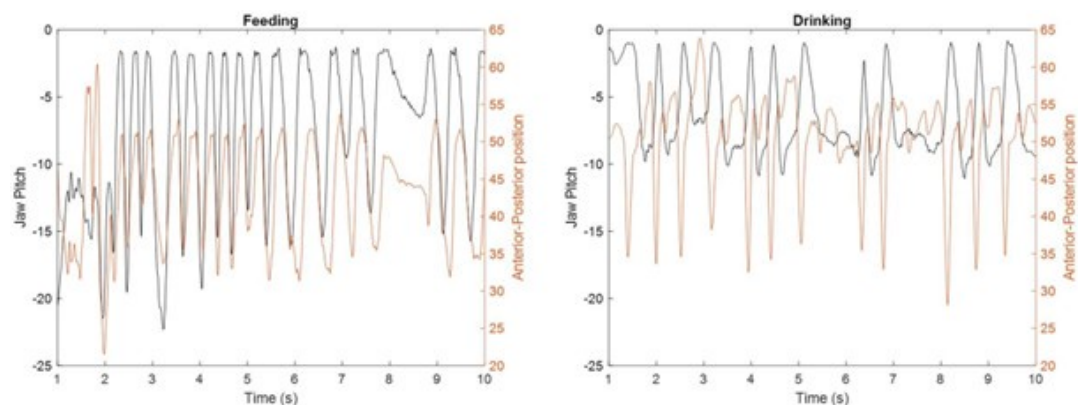
The authors compare the tuning properties during feeding to those during licking but only focus on the tongue-tip. However, the two behaviors are different also in their engagement of the jaw muscles. Thus many of the differences observed between the two 'tasks' might have very little to do with an alternation in the properties of the neural code - and more to do with the differences in the movements involved.

Using the tongue tip for the kinematic analysis of tongue directional movements was a deliberate choice as the anterior region of the tongue is highly mobile and sensitive due to a higher density of mechanoreceptors. The tongue tip is the first region that touches the spout in the drinking task and moves the food into the oral cavity for chewing and subsequent swallowing.

We agree with the reviewer that the jaw muscles are engaged differently in feeding vs. drinking (see Fig. 2). For example, a wider variety of jaw movements along the three axes are observed in feeding compared to the smaller amplitude and mostly vertical jaw movements in drinking. Also, the tongue movements are very different between the two behaviors. In feeding, the tongue moves in varied directions to position the food between left-right tooth rows during chewing, whereas in the drinking task, the tongue moves to discrete locations to receive the juice reward. Moreover, the tongue-jaw coordination differs between tasks; maximum tongue protrusion coincides with maximum gape in drinking but with minimum gape in the feeding behavior. Thus, the different tongue and jaw movements required in each behavior may account for some of the differences observed in the directional tuning properties of individual neurons and population activity. These points have been included in the revised Discussion.

Author response image 3.

Tongue tip position (mm) and jaw pitch(degree) during feeding (left) and drinking (right) behaviors. Most protruded tongue position coincides with minimum gape (jaw pitch at 0°) during feeding but with maximum gape during drinking.



Many of the neurons are likely correlated with both Jaw movements and tongue movements - this complicates the interpretations and raises the possibility that the differences in tuning properties across tasks are trivial.

We thank the reviewer for raising this important point. In fact, we verified in a previous study whether the correlation between the tongue and jaw kinematics might explain differences in the encoding of tongue kinematics and shape in M1o (see Supplementary Fig. 4 in Laurence-Chasen et al., 2023): “Through iterative sampling of sub-regions of the test trials, we found that correlation of tongue kinematic variables with mandibular motion does not account for decoding accuracy. Even at times where tongue motion was completely uncorrelated with the jaw, decoding accuracy could be quite high.”

The results obtained from population analyses showing distinct properties of population trajectories in feeding vs. drinking behaviors provide strong support to the interpretation that directional information varies between these behaviors.

The population analyses for decoding are rudimentary and provide very coarse estimates (left, center, or right), it is also unclear what the major takeaways from the population decoding analyses are. The reduced classification accuracy could very well be a consequence of linear models being unable to account for the complexity of feeding movements, while the licking movements are 'simpler' and thus are better accounted for.

We thank the reviewer for raising this point. The population decoding analyses provide additional insight on the directional information in population activity, as well as a point of comparison with the results of numerous decoding studies on the arm region of the sensorimotor cortex. In the revised version, we have included the results from decoding tongue direction using a long short-term memory (LSTM) network for sequence-to-sequence decoding. These results differed from the KNN results, indicating that a linear model such as KNN was better for drinking and that a non-linear and continuous decoder was better suited for feeding. These results have been included in the revised manuscript.

The nature of the nerve block and what sensory pathways are being affected is unclear - the trigeminal nerve contains many different sensory afferents - is there a characterization of how effectively the nerve impulses are being blocked? Have the authors confirmed or characterized the strength of their inactivation or block, I was unable to find any electrophysiological evidence characterizing the perturbation.

The strength of the nerve block is characterized by a decrease in the baseline firing rate of S1o neurons, as shown in Supplementary Figure 6 of “Loss of oral sensation impairs feeding performance and consistency of tongue–jaw coordination” (Laurence-Chasen et al., 2022)..

Overall, while this paper provides a descriptive account of the observed neural correlations and their alteration by perturbation, a synthesis of the observed changes and some insight into neural processing of tongue kinematics would strengthen this paper.

We thank the reviewer for this suggestion. We have revised the Discussion to provide a synthesis of the results and insights into the neural processing of tongue kinematics.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) The procedure for anesthesia explained in the method section was not clear to me. The following information was missing: what drug/dose was used? How long the animal was under anesthesia? How long after the recovery the experiments were done?

The animals were fully sedated with ketamine (100 mg/ml, 10 mg/kg) for less than 30 minutes, and all of the data was collected within 90 minutes after the nerve block was administered.

(2) In Figure 10, panels A and B are very close together, it was not at first clear whether the text "Monkey R, Monkey Y" belongs to panel A or B.

We have separated the two panels further in the revised figure.

(3) I found Figure 11 very busy and hard to interpret. Separating monkeys, fitting the line for each condition, or using a bar plot can help with the readability of the figure.

Thank you for the suggestion. We agree with you and have reworked this figure. To simplify it we have shown the mean accuracy across iterations.

(4) I found the laterality discussions like "This signifies that there are more neurons in the left hemisphere contributes toward one direction of tongue movement, suggesting that there is some laterality in the PDs of OSMCx neurons that varies between individuals" bit of an over-interpretation of data, given the low n value and the dissimilarity in how strongly the nerve blocking altered monkeys behavior.

Thank you for sharing this viewpoint. We do think that laterality is a good point of comparison with studies on M1 neurons in the arm/hand region. In our study, we found that the peak of the PD distribution coincides with leftward tongue movements in feeding. The distribution of PDs provides insight into how tongue muscles are coordinated during movement. Intrinsic and extrinsic tongue muscles are involved in shaping the tongue (e.g., elongation, broadening) and positioning the tongue (e.g., protrusion/retraction, elevation/depression), respectively. These muscles receive bilateral motor innervation except for genioglossus. Straight tongue protrusion requires the balanced action of the right and left genioglossi while the lateral protrusion involves primarily the contralateral genioglossus. Given this unilateral innervation pattern, we hypothesized that left MIO/SIO neurons would preferentially respond to leftward tongue movements, corresponding to right genioglossus activation.

Reviewer #2 (Recommendations for the authors):

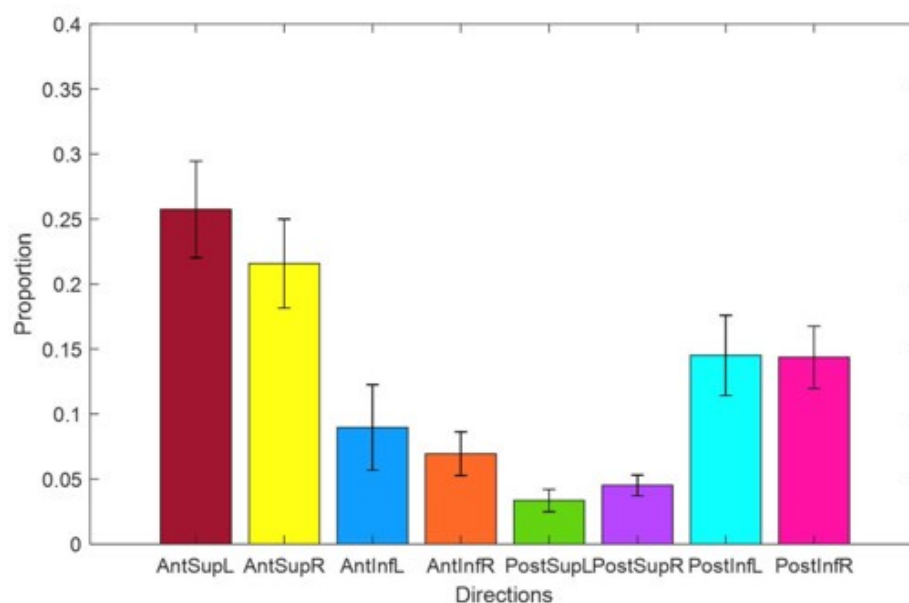
Are the observation of tuning peaks being most frequently observed toward the anterior and superior directions consistent with the statistics of the movements the tongue typically makes? This could be analogous to anisotropies previously reported in the arm

literature, e.g., Lillicrap TP, Scott SH. 2013. Preference Distributions of Primary Motor Cortex Neurons Reflect Control Solutions Optimized for Limb Biomechanics. *Neuron*. 77(1):168-79

Thank you for bringing our attention to analogous findings by Lillicrap & Scott, 2013. Indeed, we do observe the highest number of movements in the Anterior Superior directions, followed by the Posterior Inferior. This does align with the distribution of tuning peaks that we observed. Author response image 4 shows the proportions of observed movements in each group of directions across all feeding datasets. We have incorporated this data in the Results section: Neuronal modulation patterns differ between MIO and SIO, as well as added this point in the Discussion.

Author response image 4.

Proportion of feeding trials in each group of directions. Error bars represent ± 1 standard deviation across datasets ($n = 4$).



"The Euclidean distance was used to identify nearest neighbors, and the number of nearest neighbors used was $K = 7$. This K value was determined after testing different K s which yielded comparable results." In general, it's a decoding best practice to tune hyperparameters (like K) on fully held-out data from the data used for evaluation. Otherwise, this tends to slightly inflate performance because one picks the hyperparameter that happened to give the best result. It sounds like that held-out validation set wasn't used here. I don't think that's going to change the results much at all (especially given the "comparable results" comment), but providing this suggestion for the future. If the authors replicate results on other datasets, I suggest they keep $K = 7$ to lock in the method.

$K = 7$ was chosen based on the size of our smallest training dataset ($n = 55$). The purpose of testing different K values was not to select which value gave the best result, but to demonstrate that similar K values did not affect the results significantly. We tested the different K values on a subset of the feeding data, but that data was not fully held-out from the training set. We will keep your suggestion in mind for future analysis.

The smoothing applied to Figure 2 PSTHs appears perhaps excessive (i.e., it may be obscuring interesting finer-grained details of these fast movements). Can the authors reduce the 50 ms Gaussian smoothing (I assume this is the s.d.?) ~25 ms is often used in studying arm kinematics. It also looks like the movement-related modulation may not be finished in these 200 ms / 500 ms windows. I suggest extending the shown time window. It would also be helpful to show some trial-averaged behavior (e.g. speed or % displacement from start) under or behind the PSTHs, to give a sense of what phase of the movement the neural activity corresponds to.

Thank you for the suggestion. We have taken your suggestions into consideration and modified Figure 2 accordingly. We decreased the Gaussian kernel to 25 ms and extended the time window shown. The trial-averaged anterior/posterior displacement was also added to the drinking PSTHs.

Reviewer #3 (Recommendations for the authors):

The major consideration here is that the data reported for feeding appears to be very similar to that reported in a previous study:

"Robust cortical encoding of 3D tongue shape during feeding in macaques"

Are the neurons reported here the same as the ones used in this previous paper? It is deeply concerning that this is not reported anywhere in the methods section.

These are the same neurons as in our previous paper, though here we include several additional datasets of the nerve block and drinking sessions. We have now included this in the methods section.

Second, I strongly recommend that the authors consider a thorough rewrite of this manuscript and improve the presentation of the figures. As written, it was not easy to follow the paper, the logic of the experiments, or the specific data being presented in the figures.

Thank you for this suggestion. We have done an extensive rewrite of the manuscript and revision of the figures.

A few recommendations:

(1) Please structure your results sections and use descriptive topic sentences to focus the reader. In the current version, it is unclear what the major point being conveyed for each analysis is.

Thank you for this suggestion. We have added topic sentences to the begin each section of the results.

(2) Please show raster plots for at least a few example neurons so that the readers have a sense of what the neural responses look like across trials. Is all of Figure 2 one example neuron or are they different neurons? Error bars for PETH would be useful to show the reliability and robustness of the tuning.

Figure 2 shows different neurons, one from MIO and one from SIO for each task. There is shading showing ± 1 standard error around the line for each direction, however this was a bit difficult to see. In addition to the other changes we have made to these figures, we made the

lines smaller and darkened the error bar shading to accentuate this. We also added raster plots corresponding to the same neurons represented in Figure 2 as a supplement.

(3) Since there are only two data points, I am not sure I understand why the authors have bar graphs and error bars for graphs such as Figure 3B, Figure 5B, etc. How can one have an error bar and means with just 2 data points?

Those bars represent the standard error of the proportion. We have changed the y-axis label on these figures to make this clearer.

(4) Results in Figure 6 could be due to differential placement of the electrodes across the animals. How is this being accounted for?

Yes, this is a possibility which we have mentioned in the discussion. Even with careful placement there is no guarantee to capture a set of neurons with the exact same function in two subjects, as every individual is different. Rather we focus on analyses of data within the same animal. The purpose of Figure 6 is to show the difference between MIO and SIO, and between the two tasks, within the same subject. The more salient result from calculating the preferred direction is that there is a change in the distribution between control and nerve block within the same exact population. Discussions relating to the comparison between individuals are speculative and cannot be confirmed without the inclusion of many more subjects.

(5) For Figure 7, I would recommend showing the results of the Sham injection in the same figure instead of a supplement.

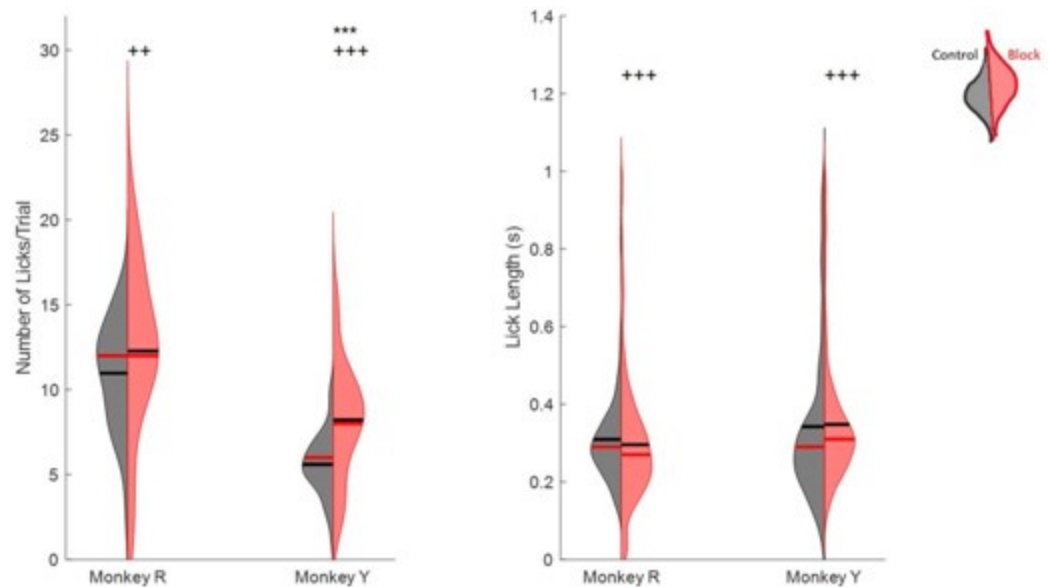
Thank you for the suggestion, we have added these results to the figure.

(6) I think the effects of the sensory block on the tongue kinematics are underexplored in Figure 7 and Figure 8. The authors could explore the deficits in tongue shape, and the temporal components of the trajectory.

Some of these effects on feeding have been explored in a previous paper, LaurenceChasen et al., 2022. We performed some additional analyses on changes to kinematics during drinking, including the number of licks per 10 second trial and the length of individual licks. The results of these are included below. We also calculated the difference in the speed of tongue movement during drinking, which generally decreased and exhibited an increase in variance with nerve block (f-test, $p < 0.001$). However, we have not included these figures in the main paper as they do not inform us about directionality.

Author response image 5.

Left halves of hemi-violins (black) are control and right halves (red) are nerve block for an individual. Horizontal black lines represent the mean and horizontal red lines the median. Results of two-tailed t-test and f-test are indicated by asterisks and crosses, respectively: *,† $p < 0.05$; **,†† $p < 0.01$; ***,††† $p < 0.001$.



(9) In Figures 9 and 10. Are the same neurons being recorded before and after the nerve block? It is unclear if the overall "population" properties are different, or if the properties of individual neurons are changing due to the nerve block.

Yes, the same neurons are being recorded before and after nerve block. Specifically, Figure 9B shows that the properties of many individual neurons do change due to the nerve block. Differences in the overall population response may be attributed to some of the units having reduced/no activity during the nerve block session.

Additionally, I recommend that the authors improve their introduction and provide more context to their discussion. Please elaborate on what you think are the main conceptual advances in your study, and place them in the context of the existing literature. By my count, there are 26 citations in this paper, 4 of which are self-citations - clearly, this can be improved upon.

Thank you for this suggestion. We have done an extensive rewrite of the Introduction and Discussion. We discussed the main conceptual advances in our study and place them in the context of the existing literature.

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